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Systems and methods for managing BYOD devices in a video conference

Abstract

Various embodiments provide systems and methods for authentication and feature enablement for bring your own device (BYOD) users of a shared conference room. For instance, when a BYOD device connects with a conference room host device, the BYOD device may communicate with a cloud-based management service to get authenticated, and the cloud-based management service may communicate and allowed or disallowed status to the conference room host device. Assuming that the BYOD device receives an allowed status, the conference room host device may then enable data communication, peripheral access, and a set of entitlements for the BYOD device.

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Background/Summary

FIELD

(1) The present disclosure is generally directed to authenticating devices in a conference room and, more specifically, to authenticating and providing service entitlements to user devices in a conference room.

BACKGROUND

(2) As the value and use of information continue to increase, individuals and businesses seek additional ways to process and store it. One option available to users is an Information Handling System (IHS). An IHS generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information.

(3) Because technology and information handling needs and requirements vary between different users or applications, IHSs may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated.
(4) Variations in IHSs allow for IHSs to be general or configured for a specific user or specific use, such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, IHSs may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.
SUMMARY
(5) Systems and methods for authenticating user devices and providing service entitlements to those user devices are described.
(6) In an illustrative, non-limiting embodiment, a method includes: identifying that a user device has connected to a conference room host device; blocking the user device from a data channel between a management agent at the conference room host device and the user device in response to identifying that the user device has connected; receiving a message from a cloud-based management service to the management agent indicating an allowed status of the user device; and opening the data channel between the user device and the management agent in response to receiving the message indicating the allowed status.
(7) In an illustrative, non-limiting embodiment, an Information Handling System (IHS) includes one or more processors; and a memory coupled to the one or more processors, wherein the memory comprises computer-readable instructions, which upon execution by the one or more processors, causes the IHS to: receive a notification from a user device, the notification indicating that the user device has connected to a conference room host device; receive user device details from the user device in response to a request for the user device details; authenticate the user device based at least in part on the user device details; and transmit a message to the conference room host device indicating an allowed status of the user device.
(8) In yet another illustrative, non-limiting embodiment, a non-transitory computer-readable storage device has instructions stored thereon for user device authentication, wherein execution of the instructions by one or more processors of an IHS (Information Handling System) causes the one or more processors to: receive a notification from a user device, the notification indicating that the user device has connected to a conference room host device; receive user device details from the user device in response to a request for the user device details; authenticate the user device based at least in part on the user device details; and transmit a message to the conference room host device indicating an allowed status of the user device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention(s) is/are illustrated by way of example and is/are not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity, and have not necessarily been drawn to scale.
(2) FIG. **1** is a diagram illustrating an example of a remote conference or meeting room, according to some embodiments.
(3) FIG. **2** is a diagram illustrating examples of components of an Information Handling System (IHS), according to some embodiments.
(4) FIG. **3** is a diagram illustrating examples of components of a video bar, according to some embodiments.
(5) FIGS. **4**A-C are diagrams illustrating examples of multi-point positioning systems and

techniques, according to some embodiments.
(6) FIG. **5** is a diagram illustrating an example of an architecture usable for determining a device's location, position, and/or orientation, according to some embodiments.
(7) FIG. **6** is a diagram illustrating an example of a method for determining a device's location, position, and/or orientation, according to some embodiments.
(8) FIG. **7** is a diagram illustrating an example of an architecture usable for handling conference room boundaries and/or context, according to some embodiments.
(9) FIGS. **8**A and **8**B are diagrams illustrating examples of methods for handling conference room boundaries and/or context, according to some embodiments.
(10) FIGS. **9**A and **9**B are diagrams illustrating examples of methods for handling conference room boundaries and/or context when a microphone is muted, according to some embodiments.
(11) FIG. **10** is a diagram illustrating an example of an architecture usable for securely adding devices to a conference room, according to some embodiments.
(12) FIG. **11** is a diagram illustrating an example of a method for securely adding devices to a conference room, according to some embodiments.
(13) FIG. **12** is a diagram illustrating an example of an architecture usable for identifying and authenticating users in a conference room, according to some embodiments.
(14) FIG. **13** is a diagram illustrating an example of a method for identifying and authenticating users in a conference room, according to some embodiments.
(15) FIG. **14** is a diagram illustrating an example of an architecture usable for BYOD devices in a conference room, according to some embodiments.
(16) FIG. **15** is a diagram illustrating an example of a method for managing BYOD devices in a conference room, according to some embodiments.
(17) FIGS. **16**A-B are a diagram illustrating an example of a method for verifying BYOD devices in a conference room, according to some embodiments.

DETAILED DESCRIPTION

(18) For purposes of this disclosure, an Information Handling System (IHS) may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an IHS may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., Personal Digital Assistant (PDA) or smartphone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price.
(19) An IHS may include Random Access Memory (RAM), one or more processing resources such as a Central Processing Unit (CPU) or hardware or software control logic, Read-Only Memory (ROM), and/or other types of nonvolatile memory. Additional components of an IHS may include one or more disk drives, one or more network ports for communicating with external devices as well as various I/O devices, such as a keyboard, a mouse, a touchscreen, and/or a video display. An IHS may also include one or more buses operable to transmit communications between the various hardware components.
(20) As used herein, the terms “heterogenous computing platform,” “heterogenous processor,” or “heterogenous platform,” and other like terms, as used herein, generally refer to various types of Integrated Circuit (ICs) or chips (e.g., a System-On-Chip or “SoC,” a Field-Programmable Gate Array or “FPGA,” an Application-Specific Integrated Circuit or “ASIC,” etc.) containing a plurality of discrete processing circuits or semiconductor Intellectual Property (IP) cores (collectively referred to as “SoC devices” or simply “devices”) in a single electronic or semiconductor package, where each device may have different processing capabilities suitable for handling a corresponding type of computational task. Examples of heterogeneous processors include, but are not limited to: QUALCOMM's SNAPDRAGON, SAMSUNG's EXYNOS,

APPLE's “A” SERIES, etc.
(21) The terms “conferencing session,” “collaboration session,” “remote conferencing,” “web conferencing,” “teleconferencing,” “videoconferencing,” “remote communication,” “remote collaboration,” “virtual collaboration,” “virtual meeting,” “remote meeting,” and other like terms, as used herein, generally refer to various types of electronic meeting, conferencing, or collaborative interactions among clients, users, or employees (collectively referred to as “users” or “IHS users”). These interactions may include, but are not limited to: meetings, broadcasts, training events, lectures, presentations, etc. (collectively referred to as “remote meetings”).
(22) In modern implementations, a remote meeting may employ several different technologies, including Unified Communication (UC) applications and services (e.g., ZOOM, TEAMS, SKYPE, FACETIME, etc.), robust (and/or lightweight) protocols, data encryption and compression techniques, etc., to enable the exchange of streams of text messages, voice, video, and/or other electronic data (e.g., files, documents, etc.) in real-time among remote users in dispersed locations.
(23) When at work, for example, a client, user, or employee (collectively referred to as “user” or “IHS user”) may participate in a remote meeting from their desk. Alternatively, the user may participate in a remote meeting from a conference room (e.g., in an office building). For example, a user may travel to the conference room carrying the IHS they intend to use. Upon arrival at the conference room, the user may also find multiple resources or devices (e.g., large displays, high-resolution cameras, whiteboards, etc.) available to them as well to any other local participants of the remote meeting.
(24) Furthermore, “hoteling,” “office hoteling,” “shared workspaces,” and “co-working spaces” are examples of environments where users schedule their hourly, daily, or weekly use of individual workspaces, such as office desks, cubicles, or conference rooms, as an alternative to permanently assigned seating. Users can access a reservation system to book an individual workspace before they arrive at work, which gives them freedom and flexibility. In some cases, individual workspaces may also be available to users without a reservation system on a first-come, first-serve basis (e.g., in the lobby of a hotel). While a user spends their allotted time in an individual workspace, they may also participate in one or more remote meetings.
(25) FIG. **1** is a diagram illustrating an example of remote conference or meeting room **100**, or any workspace equipped with similar remote meeting capabilities. In some cases, remote conference or meeting room **100** may have multiple intended uses, and it may be accessible to users on a first-come-first-serve basis without users having to schedule it through a reservation system. Alternatively, remote conference or meeting room **100** may be exclusively dedicated to remote meetings and may be accessible by reservation only.
(26) As shown, remote conference or meeting room **100** includes video bar **101** coupled to shared or external display(s) **102**, such that users may sit around table **103** and operate their respective IHSs **104**A-N (sometimes referred to as “client devices” or client IHSs) to conduct and/or participate in remote conferences or meetings. It should be noted, however that the exact configuration of conference or meeting room **100** shown in FIG. **1** may include many variations.
(27) For example, in some cases, video bar **101** may not be located adjacent to shared or external display(s) **102** (e.g., on table **103**, another wall of room **100**, etc.). In other cases, two or more shared or external displays **102** may be provided. Yet in other cases, speakers and microphones may be distributed across room **100**.
(28) Video bar **101** may include a conference or video camera, audio speakers, and/or microphones, typically housed within a single enclosure. In some implementations, video bar **101** may include an SoC, or the like, with computing capabilities that allow it to be used as an all-in-one solution for videoconferencing. To that end, video bar **101** may be configured with any software application that supports video and audio drivers, including UC applications.
(29) In certain implementations, however, video bar **101** may be coupled to an in-room or host IHS (e.g., device **105**) configured to support its operations and/or perform at least a portion of the

computations not directly performed by video bar **101** in order to enable remote meetings.
(30) One or more of IHSs **104**A-N may be coupled via cable **106** (or wirelessly) to in-room device **105**. An IHS (e.g., IHS **104**A) coupled to device **105** may make use of shared or external display(s) **102** in addition to, or as an alternative to, that IHS's integrated display.
(31) In-room device **105** may include a touch controller, in-room or host IHS, device hub, or the like. For instance, when in-room device **105** includes a touch controller (e.g., an IHS with a table form factor), it may be used by any user of IHSs **104**A-N to start and end a remote meeting, to modify certain application settings (e.g., enable screen sharing or text transcript), to reserve additional meeting time, to manage recordings and documents, to upgrade software or firmware, to add or remove peripherals, and so on.
(32) To that end, video bar **101** (and/or host IHS **105**A) may be configured to implement, execute, or instantiate an On-the-Box (OTB) agent configured to interface with a UC application or service during a remote meeting. Meanwhile, each of IHSs **104**A-N may be configured to implement, execute, or instantiate a respective client IHS agent configured to interface with the UC application (or its local instance of it) or service during the remote meeting.
(33) In various implementations, systems and methods described herein may be executed, at least in part, through interactions between the OTB agent and one or more client IHS agents. In some cases, these interactions between among the OTB agent and one or more client IHS agents may be supported by, or involve the cooperation of, one or more cloud-based services.
(34) In operation, video bar **101** and/or in-room IHS **105** may be configured to receive images obtained with one or more camera(s) located in room **100** and to share those images with remote participants of a remote meeting (e.g., as live-video images) using the UC application and service. Additionally, or alternatively, video bar **101** and/or in-room IHS **105** may be configured to receive images from remote participants (or shared documents, etc.) and to display those images to local participants of the remote meeting in on one or more displays(s) in room **100** using the UC application and service.
(35) Video bar **101** and/or in-room IHS **105** may also be configured to receive audio captured with one or more microphones(s) located in room **100** and to share that audio with remote participants of the remote meeting using the UC application and service. Video bar **101** and/or in-room IHS **105** may be further configured to receive audio captured from remote participants and to reproduce that audio to local participants of the remote meeting via one or more speakers(s) in room **100** using the UC application and service. Moreover, video bar **101** and/or in-room IHS **105** may be configured to receive a screen capture from a digital whiteboard and to share that screen capture with remote participants of the remote meeting using the UC application and service.
(36) FIG. **2** is a block diagram of components of IHS **200**. In various embodiments, IHS **200** may be used to implement aspects of IHSs **104**A-N, video bar **101**, and/or in-room device **105**. As depicted, IHS **200** includes host processor(s) **201**.
(37) IHS **200** may be a single-processor system or a multi-processor system including two or more processors. Host processor(s) **201** may include any processor capable of executing program instructions, such as an INTEL/AMD x86 processor, or any general-purpose or embedded processor implementing any of a variety of Instruction Set Architectures (ISAs), such as a Complex Instruction Set Computer (CISC) ISA, a Reduced Instruction Set Computer (RISC) ISA (e.g., one or more ARM core(s), or the like).
(38) IHS **200** includes chipset **202** coupled to host processor(s) **201**. Chipset **202** may provide host processor(s) **201** with access to several resources. In some cases, chipset **202** may utilize a QuickPath Interconnect (QPI) bus to communicate with host processor(s) **201**. Chipset **202** may also be coupled to communication interface(s) **205** to enable communications between IHS **200** and various wired and/or wireless devices or networks, such as Ethernet, WiFi, BLUETOOTH (BT), BT Low-Energy (BLE), cellular or mobile networks (e.g., Code-Division Multiple Access or “CDMA,” Time-Division Multiple Access or “TDMA,” Long-Term Evolution or “LTE,” etc.),

satellite networks, or the like.
(39) Communication interface(s) **205** may be used to communicate with peripherals devices (e.g., BT speakers, microphones, headsets, etc.). Moreover, communication interface(s) **205** may be coupled to chipset **202** via a Peripheral Component Interconnect Express (PCIe) bus, or the like.
(40) Chipset **202** may be coupled to display and/or touchscreen controller(s) **204**, which may include one or more or Graphics Processor Units (GPUs) on a graphics bus, such as an Accelerated Graphics Port (AGP) or PCIe bus. As shown, display controller(s) **204** provide video or display signals to one or more display device(s) **211**.
(41) Display device(s) **211** may include Liquid Crystal Display (LCD), light-emitting diode (LED), organic LED (OLED), or other thin film display technologies. Display device(s) **211** may include a plurality of pixels arranged in a matrix, configured to display visual information, such as text, two-dimensional images, video, three-dimensional images, etc. In some cases, display device(s) **211** may be provided as a single continuous display, rather than two discrete displays.
(42) Chipset **202** may provide host processor(s) **201** and/or display controller(s) **204** with access to system memory **203**. In various implementations, system memory **203** may be implemented using any other suitable memory technology, such as static RAM (SRAM), dynamic RAM (DRAM) or magnetic disks, or any nonvolatile/Flash-type memory, such as a Solid-State Drive (SSD), Non-Volatile Memory Express (NVMe), or the like.
(43) In certain embodiments, chipset **202** may also provide host processor(s) **201** with access to one or more Universal Serial Bus (USB) ports **208**, to which one or more peripheral devices may be coupled (e.g., integrated or external webcams, microphones, speakers, etc.).
(44) Chipset **202** may further provide host processor(s) **201** with access to one or more hard disk drives, solid-state drives, optical drives, or other removable-media drives **213**.
(45) Chipset **202** may also provide access to one or more user input devices **206**, for example, using a super I/O controller or the like. Examples of user input devices **206** include, but are not limited to, microphone(s) **214**A, camera(s) **214**B, and keyboard/mouse **214**N. Other user input devices **206** may include a touchpad, stylus or active pen, totem, etc. Each of user input devices **206** may include a respective controller (e.g., a touchpad may have its own touchpad controller) that interfaces with chipset **202** through a wired or wireless connection (e.g., via communication interfaces(s) **205**). In some cases, chipset **202** may also provide access to one or more user output devices (e.g., video projectors, paper printers, 3D printers, loudspeakers, audio headsets, Virtual/Augmented Reality (VR/AR) devices, etc.).
(46) In certain implementations, chipset **202** may provide an interface for communications with one or more hardware sensors **220**. Sensors **220** may be disposed on or within the chassis of IHS **200**, or otherwise coupled to IHS **200**, and may include, but are not limited to, electric, magnetic, radio, optical (e.g., camera, webcam, etc.), infrared, thermal, force, pressure, acoustic (e.g., microphone), ultrasonic, proximity, position, deformation, bending, direction, movement, velocity, rotation, gyroscope, Inertial Measurement Unit (IMU), and/or acceleration sensor(s).
(47) BIOS **207** is coupled to chipset **202**. UEFI was designed as a successor to BIOS, and many modern IHSs utilize UEFI in addition to or instead of the BIOS. Accordingly, BIOS/UEFI **207** is intended to also encompass corresponding UEFI component(s). BIOS/UEFI **207** provides an abstraction layer that allows the OS to interface with certain hardware components that are utilized by IHS **200**.
(48) Upon booting of IHS **200**, host processor(s) **201** may utilize program instructions of BIOS **207** to initialize and test hardware components coupled to IHS **200**, and to load host OS **300** for use by IHS **200**. Via the hardware abstraction layer provided by BIOS/UEFI **207**, software stored in system memory **203** and executed by host processor(s) **201** can interface with certain I/O devices that are coupled to IHS **200**.
(49) Embedded Controller (EC) **209** (sometimes referred to as a Baseboard Management Controller or “BMC”) includes a microcontroller unit or processing core dedicated to handling selected IHS

operations not ordinarily handled by host processor(s) **201**. Examples of such operations may include, but are not limited to, power sequencing, power management, receiving and processing signals from a keyboard or touchpad, as well as other buttons and switches (e.g., power button, laptop lid switch, etc.), receiving and processing thermal measurements (e.g., performing cooling fan control, CPU and GPU throttling, and emergency shutdown), controlling indicator Light-Emitting Diodes or "LEDs" (e.g., caps lock, scroll lock, num lock, battery, ac, power, wireless LAN, sleep, etc.), managing the battery charger and the battery, enabling remote management, diagnostics, and remediation over network(s) **203**, etc.
(50) Unlike other devices in IHS **200**, EC **209** may be made operational from the very start of each power reset, before other devices are fully running or powered on. As such, EC **209** may be responsible for interfacing with a power adapter to manage the power consumption of IHS **200**. These operations may be utilized to determine the power status of IHS **200**, such as whether IHS **200** is operating from battery power or is plugged into an AC power source. Firmware instructions utilized by EC **209** may be used to manage other core operations of IHS **200** (e.g., turbo modes, maximum operating clock frequencies of certain components, etc.).
(51) In some cases, EC **209** may implement operations for detecting certain changes to the physical configuration or posture of IHS **200** and managing other devices in different configurations of IHS **200**. For instance, when IHS **200** has a 2-in-1 laptop/tablet form factor, EC **209** may receive inputs from a lid position or hinge angle sensor **220**, and it may use those inputs to determine: whether the two sides of IHS **200** have been latched together to a closed position or a tablet position, the magnitude of a hinge or lid angle, etc. In response to these changes, the EC may enable or disable certain features of IHS **200** (e.g., front or rear-facing camera, etc.).
(52) In this manner, EC **209** may identify any number of IHS postures, including, but not limited to: laptop, stand, tablet, or book. For example, when display(s) **211** of IHS **200** is open with respect to a horizontal keyboard portion, and the keyboard is facing up, EC **209** may determine IHS **200** to be in a laptop posture. When display(s) **211** of IHS **200** is open with respect to the horizontal keyboard portion, but the keyboard is facing down (e.g., its keys are against the top surface of a table), EC **209** may determine IHS **200** to be in its stand posture. When the back of display(s) **211** is closed against the back of the keyboard portion, EC **209** may determine IHS **200** to be in a tablet posture. When IHS **200** has two display(s) **211** open side-by-side, EC **209** may determine IHS **200** to be in a book posture. In some implementations, EC **209** may also determine if display(s) **211** of IHS **200** are in a landscape or portrait orientation.
(53) In some implementations, EC **209** may be installed as a Trusted Execution Environment (TEE) component to the motherboard of IHS **200**. Additionally, or alternatively, EC **209** may be further configured to calculate hashes or signatures that uniquely identify individual components of IHS **200**. In such scenarios, EC **209** may calculate a hash value based on the configuration of a hardware and/or software component coupled to IHS **200**. For instance, EC **209** may calculate a hash value based on all firmware and other code or settings stored in an onboard memory of a hardware component.
(54) Hash values may be calculated as part of a trusted process of manufacturing IHS **200** and may be maintained in secure storage as a reference signature. EC **209** may later recalculate the hash value for a component and may compare it against the reference hash value to determine if any modifications have been made to the component, thus indicating that the component has been compromised. As such, EC **209** may validate the integrity of hardware and software components installed in IHS **200**.
(55) In addition, EC **209** may provide an OOB channel that allows an Information Technology Decision Maker (ITDM) or Original Equipment Manufacturer (OEM) to manage IHS **200**'s various settings and configurations, for example, by issuing Out-of-Band (OOB) commands.
(56) In various embodiments, IHS **200** may be coupled to an external power source through an AC adapter, power brick, or the like. The AC adapter may be removably coupled to a battery charge

controller to provide IHS **200** with a source of DC power provided by battery cells of a battery system in the form of a battery pack (e.g., a lithium-ion or “Li-ion” battery pack, or a nickel metal hydride or “NiMH” battery pack including one or more rechargeable batteries).

(57) Battery Management Unit (BMU) **212** may be coupled to EC **209** and it may include, for example, an Analog Front End (AFE), storage (e.g., non-volatile memory), and a microcontroller. In some cases, BMU **212** may be configured to collect and store information, and to provide that information to other IHS components, such as for example, devices within heterogeneous computing platform **300** (FIG. **3**).

(58) Examples of information collectible by BMU **212** may include, but are not limited to: operating conditions (e.g., battery operating conditions including battery state information such as battery current amplitude and/or current direction, battery voltage, battery charge cycles, battery state of charge, battery state of health, battery temperature, battery usage data such as charging and discharging data; and/or IHS operating conditions such as processor operating speed data, system power management and cooling system settings, state of “system present” pin signal), environmental or contextual information (e.g., such as ambient temperature, relative humidity, system geolocation measured by GPS or triangulation, time and date, etc.), events, etc.

(59) Examples of events may include, but are not limited to: acceleration or shock events, system transportation events, exposure to elevated temperature for extended time periods, high discharge current rate, combinations of battery voltage, battery current and/or battery temperature (e.g., elevated temperature event at full charge and/or high voltage causes more battery degradation than lower voltage), etc.

(60) In some embodiments, IHS **200** may not include all the components shown in FIG. **2**. In other embodiments, IHS **200** may include other components in addition to those that are shown in FIG. **2**. Furthermore, some components that are represented as separate components in FIG. **2** may instead be integrated with other components, such that all or a portion of the operations executed by the illustrated components may instead be executed by the integrated component.

(61) For example, in various embodiments described herein, host processor(s) **201** and/or other components shown in FIG. **2** (e.g., chipset **202**, display controller(s) **204**, communication interface(s) **205**, EC **209**, etc.) may be replaced by devices within heterogenous computing platform **300** (FIG. **3**). As such, IHS **200** may assume different form factors including, but not limited to: servers, workstations, desktops, laptops, appliances, video game consoles, tablets, smartphones, video bars, etc.

(62) FIG. **3** is a diagram illustrating examples of components **300** of video bar **101**. As shown, video bar **101** includes SoC **301**, such as a heterogeneous computing platform, or the like. Video bar **101** may also include audio controller **302**, video controller **303**, camera controller **304**, and communication interface(s) **305**.

(63) In some implementations, one or more of components **302-305** may be integrated directly into SoC **301**. Such integrated components or “IP cores” may be coupled to one more processing cores of SoC **301** via an interconnect fabric, or the like. In other cases, one or more of components **302-305** may be external to SoC **301**, and may be coupled to the one more processing cores via a bus, or the like.

(64) Communication interface(s) **305** may enable communications between video bar **101** and various wired and/or wireless networks, such as Ethernet, WiFi, BT/BLE, cellular or mobile networks, satellite networks, or the like. Communication interface(s) **305** may also enable communications between video bar **101** and shared or external display(s) **102** (e.g., via a Video Graphics Array or “VGA” interface, a High-Definition Multimedia Interface or “HDMI” interface, etc.). In addition, communication interface(s) **305** may enable communications with USB devices or the like. As such, communication interface(s) **305** may be used to enable various types of communications between video bar **101** and wired/wired networks, the Internet, other IHSs, BT speakers, microphones, headsets, external displays, touch controllers, whiteboards, hard drives,

peripherals, etc.
(65) Video bar **101** is coupled to shared or external display(s) **102**A-N, microphone(s) **306**, speaker(s) **307** and camera(s) **308**. Video bar **101** may also include wireless antenna(s) **309**, which may be coupled to communication interface(s) **305**.
(66) In various embodiments, video bar **101** may be equipped with multi-point positioning technology that enables it to determine the presence, location, and/or orientation of IHSs **104**A-N (or of an integrated display of IHSs **104**A-N) within room **100**. For example, high-accuracy distance measurements may be performed using phase-based ranging protocols, or the like.
(67) Video bar **101** is further coupled to host IHS **105**A and in-room touch controller **105**B (e.g., touchscreen controller(s) **204**). In some implementations, host IHS **105**A and/or touch controller **105**B may facilitate the operation of video bar **101**, or aspects thereof, in meeting room **100**, and may be coupled to in-room peripherals **310** (e.g., shared displays, whiteboards, microphones, speakers, lighting systems, HVAC controller or thermostat, etc.).
(68) In some implementations, components **105**A and/or **105**B may be absent, such as when SoC **301** is equipped with resources sufficient to perform advanced operations such as, for example, compute-intensive Artificial Intelligence (AI) or Machine Learning (ML) operations (e.g., gesture or facial recognition, etc.), encryption and decryption algorithms, etc., without the need for additional computing power. In such cases, SoC **301** may include a high-performance AI device such as a Neural Processing Unit (NPU), a Tensor Processing Unit (TSU), a Neural Network Processor (NNP), or an Intelligence Processing Unit (IPU), and it may be designed specifically for AI/ML, which speeds up the processing of AI/ML tasks.
(69) In various embodiments, SoC **301** may be configured to execute one or more AI/ML model(s). Such AI/ML model(s) may implement: a neural network (e.g., artificial neural network, deep neural network, convolutional neural network, recurrent neural network, autoencoders, reinforcement learning, etc.), fuzzy logic, deep learning, deep structured learning hierarchical learning, Support Vector Machine (SVM) (e.g., linear SVM, nonlinear SVM, SVM regression, etc.), decision tree learning (e.g., classification and regression tree or “CART”), Very Fast Decision Tree (VFDT), ensemble methods (e.g., ensemble learning, Random Forests, Bagging and Pasting, Patches and Subspaces, Boosting, Stacking, etc.), dimensionality reduction (e.g., Projection, Manifold Learning, Principal Components Analysis, etc.), or the like.
(70) Non-limiting examples of available AI/ML algorithms, models, software, and libraries that may be utilized within embodiments of systems and methods described herein include, but are not limited to: PYTHON, OPENCV, INCEPTION, THEANO, TORCH, PYTORCH, PYLEARN2, NUMPY, BLOCKS, TENSORFLOW, MXNET, CAFFE, LASAGNE, KERAS, CHAINER, MATLAB Deep Learning, CNTK, MatConvNet (a MATLAB toolbox implementing convolutional neural networks for computer vision applications), DeepLearnToolbox (a Matlab toolbox for Deep Learning from Rasmus Berg Palm), BigDL, Cuda-Convnet (a fast C++/CUDA implementation of convolutional or feed-forward neural networks), Deep Belief Networks, RNNLM, RNNLIB-RNNLIB, matrbm, deeplearning4j, Eblearn.lsh, deepmat, MShadow, Matplotlib, SciPy, CXXNET, Nengo-Nengo, Eblearn, cudamat, Gnumpy, 3-way factored RBM and mcRBM, mPoT, ConvNet, ELEKTRONN, OpenNN, NEURALDESIGNER, Theano Generalized Hebbian Learning, Apache SINGA, Lightnet, and SimpleDNN.
(71) FIGS. **4**A-C are diagrams illustrating examples of multi-point positioning systems **400**A/B and techniques **400**C, according to some embodiments. Particularly, in FIG. **4**A, multi-point positioning system **400**A includes a number of IHSs **104**A-N within range of video bar **101** in room **100**. Video bar **101** is shown as implementing, executing, or instantiating Real-Time Locating System (RTLS) node manager **401**, which is coupled to RTLS passive antenna **403** (e.g., one of antennas **309**) and RTLS host or active antenna **402**.
(72) In operation, RTLS host or active antenna **402** may be used by RTLS node manager **401** to transmit electromagnetic signal(s) or beacon(s) **404**, for example, and to receive acknowledgment

(ACK) message(s) **405**A-N from client IHSs **104**A-N in response thereto. RTLS passive antenna **403** may be used to listen to "ping" and/or acknowledgment (ACK) messages **406**A-N in parallel with RTLS host or active antenna **402**. As a result, RTLS node manager **401** may receive distance information from RTLS antennas **402** and **403** for an improved survey of devices disposed within room **100**.
(73) In operation, RTLS host or active antenna **402** may initiate a Time-of-Flight (ToF) sequence by broadcasting ToF_PING signal **404**. Client IHS **104**A may listen to RTLS host or active antenna **402** for ToF_PING signal **404** and, upon receipt, it may transmit a ToF ACK **405**A (e.g., after a selected or deterministic amount of time delay). Each of client IHSs **104**A-N may transmit its own ToF ACK **405**A-N signals back to RTLS host or active antenna **402**. Additionally, or alternatively, each ToF_PING signal **404** may be stamped with a time of transmittal, which is then compared with a time the signal is received by client IHS **104**A, and from which a ToF may be calculated. In that case, the ToF ACK signal **405**A from client IHS **104**A may include an indication of the ToF.
(74) In some implementations, a Received Signal Strength Indicator (RSSI) level of each received signal may also be used to help calculate a location and/or orientation of various devices. Furthermore, RTLS passive antenna **403** may listen for ToF_PING and ToF_ACK signals usable to calculate additional ToF values that may be used to increase spatial diversity.
(75) In some cases (e.g., large rooms, multipath scenarios, etc.), phase-based RTLS BT systems may operate using multiple frequencies, for example, as outlined in the BT 4.0 (BLE) Specifications.
(76) In other cases (e.g., small rooms), measurements using a single frequency may suffice. A flag usable as a constant tone may be sent using a connectionless (e.g., a beacon) or connected (e.g., through a data channel) protocol data unit (PDU), along with constant tone. This allows for device positioning without the need to pair client IHS **104**A to video bar **101**. Generally speaking, a positioning accuracy of 5 cm with a total measurement time of approximately 25 ms may be achieved.
(77) To that end, FIG. **4**B shows antenna system **400**B disposed within client IHS **104**A and/or integrated into IHS chassis **407**. Although shown as IHS **104**A as an example, in other cases, devices equipped with antenna system **400**B may include peripherals or other in-room devices to be located. Specifically, antenna system **400**B may include a plurality of antennas **408**A-F, such as, for example: a first WiFi antenna, a second WiFi antenna, a 5G main antenna, a first 5G MIMO antenna, a second 5G MIMO antenna, a 5G auxiliary antenna, etc.
(78) Although antenna system **400**B shows antennas **408**A-F disposed on a keyboard, wrist rest area, or trackpad surface of IHS **100**, in other embodiments one or more of antennas **408**A-F may be disposed elsewhere on the IHS's chassis, including its lateral or bottom surfaces, behind an integrated display, around its bezel, etc.
(79) Generally speaking, any of antennas in system **400**B may be used as RTLS antennas **402** and/or **403** for RTLS purposes. This feature is particularly useful, for instance, when IHS **104**A is in a closed-lid configuration. Moreover, when IHS **104**A has a form factor such that it may assume a number of different postures (e.g., laptop, book, tablet, etc.) and some of these postures may block reception or transmission by one or more antennas, other antennas may be selected or switched in for RTLS purposes (e.g., by EC **209**). In some cases, multiple antennas **408**A-F may be used simultaneously, concurrently, or sequentially to determine the orientation (e.g., angle θ) of client IHS **104**A in room **100** and/or on table **103**.
(80) In FIG. **4**C, for example, IHS **104**A includes antennas **409**A and **409**B (any of antennas **408**A-F) separated by a fixed distance "d." Reference plane **407** for direction may be determined from the location of antennas **409**A/B. In some cases, any orthogonal plane may be configured as reference plane **407** for IHS **104**A. This information may be stored along with antenna configuration information in IHS **104**A.
(81) Video bar **101** is shown (e.g., RTLS host antenna **402**). Once distances r.sub.1 and r.sub.2 are

determined (e.g., using ToF calculations) and assuming d is known, angle θ may be computed using: θ=sin.sup.−1(r.sub.2−r.sub.1/d). The calculated location and orientation information may be sent to video bar **101** and/or to client IHS **104**A, for example, with BT messaging using a custom Generic Attribute Profile (GATT).
(82) FIG. **5** is a diagram illustrating an example of architecture **500** usable for determining a device's location, position, and/or orientation (e.g., IHSs **104**A-N). As shown, architecture **500** includes certain components that are disposed in conference room **100** and others that reside in cloud **504** (e.g., servers or other IHSs that may be accessed over the Internet, and/or the software and databases that run on those servers).
(83) In this example, conference room **100** includes client IHSs **104**A-N in communication with video bar **101** and/or host IHS **105**A. Each of IHSs **104**A-N may be configured to implement, execute, or instantiate a respective client IHS agent **510**A-N. Video bar **101** and/or host IHS **105**A may also include, or otherwise be coupled to, display(s) **102**A-N, audio device(s) **306**/**207**, camera(s) **308**, digital whiteboard **509**, and touch display or controller **105**B. In other examples, however, other devices may be present in room **100** and one or more of the devices shown may be absent.
(84) Video bar **101** and/or host IHS **105**A are configured to implement, execute, or instantiate OTB agent **501** and positioning agent **502**. Positioning agent **502** may be configured to determine a distance, location, position, and/or orientation of a device (e.g., one of IHSs **104**A-N, digital whiteboard **509**, displays **102**A-N, camera(s) **308**, audio devices **306**/**307**, touch controller **105**B, etc.) in room **100**, and OTB agent **501** may be configured to communicate the device's distance, location, position, and/or orientation information to peripheral management service **503** on cloud **504**.
(85) For example, positioning agent **502** may be configured to run the positioning algorithms described previously to locate each BT-capable peripheral (e.g., microphones, laser pointers, etc.) in conference room **100**. The distance, location, and/or orientation of peripherals with respect to video bar **101** and/or host IHS **105**A is then sent to cloud **504** for processing. Positioning agent **502** may later also query location mapping service **506** on cloud **504** for the location/position of a particular resource or peripheral in room **100** that it may need in order to allow or block selected features.
(86) Examples of hardware-based features that may be allowed, blocked, or modified on a by-device, by-user, by-room, and/or by-meeting basis depending at least in part upon the distance, location, position, and/or orientation of client IHS **104**A (or any/all other client IHSs **104**B-N in room **100**) include, but are not limited to: access to or control of devices integrated into client IHS **104**A (e.g., client IHS **104**A's integrated camera, display, microphone, speakers etc.), access to or control of devices integrated into other client IHSs **104**B-N (e.g., other client IHSs **104**B-N's integrated cameras, displays, microphones, speakers etc.), and/or access to or control of devices available in room **100** (e.g., video bar, host IHS, touchscreen controller, external or shared displays or projectors, external or shared cameras, digital whiteboards, laser pointers, room lighting system, room thermostat, etc.).
(87) Meanwhile, examples of software-based features that may be allowed, blocked, or modified on a by-device, by-user, by-room, and/or by-meeting basis depending at least in part upon the distance, position, and/or orientation of client IHS **104**A (or any/all other client IHSs **104**B-N in room **100**) include, but are not limited to: starting a remote meeting, admitting or removing a participant, muting or unmuting a microphone or speaker, changing an audio input or output gain or volume, activating or deactivating a video effect (e.g., display blur, virtual background, etc.), sharing content (e.g., file, desktop, or window sharing), recording audio and/or video, changing a status of a participant of the collaboration session, viewing or producing closed caption or live transcripts, etc.
(88) In some cases, the aforementioned hardware and software-based features may be set based

upon one or more polic(ies) associated with a particular room and/or with a specific remote meeting. Such a policy (e.g., expressed as a JavaScript Object Notation or "JSON" file, an eXtensible Markup Language or "XML" file, etc.) may be enforceable, at least in part, by OTB agent **501** executed by video bar **101** and/or host IHS **105**A.

(89) Generally speaking, polic(ies) may include rules for operating, configuring, selecting settings, etc. with respect to video bar **101**, host IHS **105**A, touch controller **105**B, client IHS **104**A, a user of client IHS **104**A, client IHSs **104**A-N (e.g., number or distribution of client IHSs in a conference room), and users of the plurality of client IHSs **104**A-N, etc. In addition, polic(ies) may include rules for operating, configuring, selecting settings, etc. with respect to any peripheral device in conference room **100**, such as display(s) **102**A-N, microphone(s) **306**, speaker(s) **307**, camera(s) **308**, digital whiteboard **509**, etc.

(90) Policy rules may output commands, notifications, and settings to be performed during, in anticipation of, and/or upon termination of a remote session, for example, depending upon a number of client IHSs in room **100**, a particular client IHS's location or orientation, and/or a particular client IHS's location or orientation relative to: video bar **101**, a wall, door, stage, or window of room **100**, other client IHS(s), an in-room display, an in-room camera, a digital whiteboard, etc., or any other suitable contextual information.

(91) In some cases, one of a plurality of external cameras may be selected during a remote meeting depending upon the orientation(s) of one or more client IHSs (e.g., an average of all participants' orientations, an average of all participants' orientations where each orientation weighed proportionally by its respective participant's role in the remote meeting, etc.) in room **100**. In other cases, one of a plurality of external displays may be similarly selected depending upon the orientation(s) of one or more client IHSs in room **100**. In yet other cases, one of a plurality of audio devices (e.g., microphones) may be similarly selected depending upon the orientation(s) of one or more client IHSs in room **100**.

(92) In other cases, a policy enforceable at least in part by video bar **100** may provide that, if there is only one client IHS in room **100**, a remote meeting session should use the client IHS's integrated camera, whereas if multiple client IHSs are detected, an external camera may be employed in addition or as an alternative thereto. In cases were SoC **301** and/or host IHS **105**A are equipped with gesture recognition features, a first camera may be selected to capture a first participant's video for broadcasting it during a remote meeting and a second camera may be selected to capture the first (or a second) participant's video for gesture recognition purposes, for example, based upon the location and/or orientation of one of client IHSs **104**A-N used by a host or speaker of a remote meeting.

(93) OTB agent **501** may be responsible for communications between positioning agent **502** and cloud **504**. Once it receives coordinates, distances, angles, etc. from positioning agent **502**, OTB agent **501** sends that information to peripheral management service **503** in cloud **504**.

(94) On cloud **504**, peripheral management service **503** is coupled to device and configuration database **505** and location mapping service **506**. Particularly, peripheral management service **503** may operate as an orchestrator in cloud **504** that connects cloud services with video bar **101** and/or host IHS **105**A.

(95) Location mapping service **506** is a core service in cloud **504** and it gathers all the individual locations of IHSs, peripherals, and other resources in conference room **100** from positioning agent **502** and generates a virtual map of room **100** with each IHS/peripheral/device identifier tagged with a coordinate in room **100**. In this manner, location mapping service **506** may maintain the location of all IHSs/peripherals/devices in across all conference rooms where the architecture is deployed. Location mapping service **506** may also publish Application Programming Interfaces (APIs) that OTB agent **501** may query to determine the location of any IHS/peripheral/device having a selected identifier.

(96) Peripheral management service **503** is further coupled to authorization (AuthZ) and

authentication (AuthN) service(s) **507**. AuthZ and AuthN service(s) **507** are coupled to user database **508**.

(97) In some cases, device and configuration database **505** may include, for each IHS recorded therein, information such as, for example: serial numbers, model numbers, service tags, device capabilities, settings, configurations, firmware versions, health status, utilization data, digital certificates, public encryption keys, device telemetry data, etc. Moreover, user database **508** may include personal or unique IHS user information, such as name, identification number, current job or position, employment history, associated or enterprise-issued client IHSs (e.g., by serial number or service tags) and peripheral devices, geographic location or address, import/export or other legal restrictions, etc.

(98) In execution, OTB agent **501**, positioning agent **502**, peripheral management service **503**, device and configuration database **505**, location mapping service **506**, AuthZ and AuthN service(s) **507**, and user database **508** may be employed to perform one or more operations described with respect to FIG. **6**.

(99) FIG. **6** is a diagram illustrating an example of method **600** for determining a device's location, position, and/or orientation. In various embodiments, method **600** may be performed, at least in part, by components of architecture **500** of FIG. **5**.

(100) Method **600** begins at **601**, where video bar **101** transmits signals (e.g., BT beacons) in meeting room **100**. At **602**, video bar **101** receives ACK messages from one or more devices, such as client IHSs **104**A-N or other peripheral devices in meeting room **100** (e.g., display(s) **102**).

(101) At **603**, video bar **101** (and/or host IHS **105**A) may determine the distance, location, and/or orientation of one or more of IHSs **104**A-N or other devices in meeting room **100**, for example, using the aforementioned ToF techniques.

(102) At **604**, in response to a determination of where IHSs **104**A-N and other devices are disposed in meeting room **100**, as well as their orientation, video bar **101** may select one or more display(s) **102**A-N to be used during a remote meeting, for example, based upon a policy. Additionally, or alternatively, at **605**, still in response to the determination, video bar **101** may also select one or more camera(s) **308** to be used during the remote meeting based on the policy.

(103) For example, in response to a situation where there is a single user in meeting room **100** and the user's IHS **104**A has its lid closed, video bar **101** may enforce a policy that turns on external display **102** and uses camera **308** (as opposed to the IHS's integrated display and camera, if any). If there is more than one display in meeting room **100**, a display that is facing IHSs **104**A-N may be selected in favor of another display or camera that is behind IHSs **104**A-N, for example. If users turn around or move during the meeting, video bar **101** may enforce a policy that selects another display or camera to be used. If a user changes the posture of IHS **104**A such that the IHS's integrated camera's field-of-view (FOV) is blocked (e.g., closed against a lid or keyboard), video bar **101** may also select another camera to be used.

(104) In some cases, microphone(s) **306** and/or speaker(s) **307** may be distributed across room **100**, and may also be selected during a remote meeting in response to changes in an IHS or peripheral device's distance, location, or orientation with respect to video bar **101**.

(105) At **606**, video bar **101** and/or host IHS **105**A may create and maintain a virtual map of conference room **100**. For example, video bar **101** may maintain a table containing the room's dimensions, as well as the location coordinates of any door, window, and/or furniture (e.g., table, sofa, etc.) located in meeting room **100**. The table may also contain the identification, location coordinates, and orientation of IHSs and peripheral devices present in room **100**. The table may also contain the identification, location coordinates, and orientation of cameras, displays, speakers, and microphones in room **100**.

(106) In some embodiments, the information contained in such a table may be rendered on a Graphical User Interface (GUI) in the form of a digital or virtual map of room **100**. In some cases, the virtual map may also be overlaid upon or otherwise combined with a live or snapshot image of

room **100**.
(107) Particularly, once an image of room **100** is obtained, SoC **301** and/or camera controller **304** may perform ML/AI feature extraction operations that identify, in one or more images of shared space **301** captured by camera(s) **103**, elements such as: objects, surfaces, and shapes (e.g., doors, windows, tables, furniture, etc.), landmarks, client IHSs, human beings (including body parts such as: head, face, eyes, ears, mouth, arm, hand, fingers, fingertips, etc.), displays, microphones, speakers, digital whiteboards, cameras, etc.
(108) Examples of feature extraction techniques and algorithms usable by SoC **301** and/or camera controller **304** to identify these elements may include, but are not limited to: edge detection, corner detection, blob detection, ridge detection, scale-invariant feature transforms, thresholding, template matching, Hough transforms, etc.
(109) As such, the distance, location, position, and/or orientation of any entity in room **100**, including client IHSs **104**A-N and any detected peripheral devices using multi-point positioning system **400**A may be reconciled against a respective distance, location, position, and/or orientation of the device as determined using images from camera(s) **308**.
(110) When facial recognition is enabled, video bar **101** and/or host IHS **105**A may perform facial recognition upon participants in room **100** and match users to their respective client IHSs **104**A-N, for example, for security or productivity purposes.
(111) After **606**, control returns to **601**. In this manner, video bar **101** may continuously and/or periodically evaluate the IHSs and devices present in room **100**, as well as their distances, locations, and/or orientations, and it may use that information to select which devices to use during a remote meeting, as well as perform other actions, as described in more detail below.
(112) As such, systems and methods described herein enable video bar **101** to gather locations of resources in a room, and to generate a virtual mapping system that may be later referred to for execution of location-based features. The use of spatial diversity with RTLS passive antenna **403** improves ranging accuracy and allows for low latency multi-device collaboration room measurements with a pre-defined single frequency tone. Moreover, these systems and methods may be used to determine the orientation of client IHSs with respect to video bar **101** and/or collaboration room **100** with multi-antenna measurements on client IHSs.
(113) It should be noted that, in various embodiments, architecture **500** may be combined with components of other architectures described herein, and method **600** may be combined with operations of other methods described herein, to provide additional or alternative features.
(114) As described herein, in an illustrative, non-limiting embodiment, a video bar may include a processor and a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution by the processor, cause the video bar to: transmit a signal in a conference room; in response to the transmission, receive an acknowledgment from a client IHS in the conference room; and determine a distance between the video bar and the client IHS based, at least in part, upon a ToF calculation, where the ToF calculation is based, at least in part, upon a difference between: (i) a time the acknowledgment is received, and (ii) a time of the transmission.
(115) The processor may include or be coupled to a video camera controller configured to capture an image of at least a portion of the conference room. Additionally, or alternatively, the processor may include or be coupled to a display controller configured to render an image on a shared display during a conference session. The signal may be broadcast as a BT signal.
(116) The video bar may include an active RTLS and a passive RTLS in communication with a node manager executed by the processor. The program instructions, upon execution by the processor, cause the video bar to transmit the signal using the active RTLS and to receive the acknowledgment via the active RTLS and the passive RTLS. In some cases, the distance may be determined using a phase-based ranging technique.
(117) The program instructions, upon execution by the processor, further cause the video bar to determine a location of the client IHS in the conference room. The program instructions, upon

execution by the processor, may also cause the video bar to determine the orientation of the client IHS in the conference room.
(118) The orientation may include or indicate an orientation of a display integrated into the client IHS. In some cases, the orientation may be determined with respect to a shared display disposed in the conference room. In other cases, the orientation may be determined with respect to a camera disposed in the conference room.
(119) The program instructions, upon execution by the processor, cause the video bar to select one or more of a plurality of cameras disposed in the conference room to capture an image based, at least in part, upon the orientation. For example, the plurality of cameras may include: (i) a camera integrated into the client IHS, (ii) another camera integrated into or coupled to the video bar, and (iii) yet another camera integrated into another client IHS.
(120) The program instructions, upon execution by the processor, may also cause the video bar to select one or more of a plurality of displays disposed in the conference room to render an image based, at least in part, upon the orientation. For example, the plurality of displays may include: (i) a display integrated into the client IHS, (ii) another display coupled to the video bar, or (iii) yet another display integrated into another client IHS.
(121) In some implementations, the processor may be coupled to a host IHS external to the video bar, and the host IHS may be configured to make at least a portion of the ToF calculation or the determination. The program instructions, upon execution by the processor, may further cause the video bar or the host IHS to create or maintain a virtual map of one or more client IHSs disposed in the conference room.
(122) In another illustrative, non-limiting embodiment, a memory device may have program instructions stored thereon that, upon execution by a processor of a host IHS coupled to a video bar in a conference room, cause the host IHS to: determine distances between the video bar and a plurality of client IHSs based, at least in part, upon ToF calculations; and create or maintain a virtual map of one or more of the plurality of client IHSs disposed in the conference room during a conference session based, at least in part, upon the distances.
(123) In yet another illustrative, non-limiting embodiment, a method may include determining an orientation of each of a plurality of client IHSs disposed in the conference room, at least in part, using a video bar; and based at least in part upon the orientations, selecting at least one of: (i) one or more of a plurality of displays disposed in the conference room to render an image during a conference session, or (ii) one or more of a plurality of cameras disposed in the conference room to capture another image to be transmitted to a remote participant of the conference session.
(124) In various embodiments, systems and methods described herein may be used for handling conference room boundaries and/or context. In a wired conference room environment, users within a meeting room boundary are automatically allowed to connect and control meeting room devices on the basis of their physical presence in the room. For example, a user may control conference room devices (e.g., video bar **101**, display **102**, etc.) using physical buttons or by connecting cable **106** to IHS **104**A running configuration software.
(125) As more IHSs are deployed with wireless features, the inventors hereof have identified a need to detect meeting room boundaries and/or to prevent users from accidentally trying to connect to devices in an adjacent meeting room.
(126) Generally, if a user tries to discover and connect a wireless display or Bluetooth device of a meeting room, several issues may be observed, such as: (1) multiple devices in nearby meeting rooms are be discovered wirelessly and the user needs to rely on some naming convention to know which device to connect, and (2) when connecting for the first time, user might be prompted to key in access code displayed in the room (this is partly to prevent accidental connection to a wrong nearby device), which is a cumbersome process.
(127) Using systems and methods described herein, however, when user walks into meeting room **100**, their IHS accurately and wirelessly identifies the meeting room context, without the need for

additional mechanisms to prevent mistaken identification of other meeting rooms nearby. Once the meeting room context is identified, software on an IHS may present relevant options for its user, such as, for example: wireless control of conference room devices, automatically preventing audio feedback, collaborating with other users in the same meeting virtually, etc. Moreover, when the user walks out of meeting room **100**, the meeting room context may be automatically discarded or invalidated, along with any invalidated previously available to them.

(128) In some embodiments, to assemble meeting room context information, video bar **101** and/or host IHS **100** may combine: (i) the physical boundaries of meeting room **100**, determined using ultrasonic frequencies or other proximity-based technologies, with (ii) meeting information associated with an ongoing or upcoming remote meeting identification. For example, meeting room context may include room data (e.g., room name or ID, device list, virtual map, etc.), wireless credentials (e.g., tokens, keys, certificates, etc.) and meeting data (e.g., meeting ID, meeting app, duration, invited participants, present participants, role(s) of IHS user(s), etc.).

(129) Ultrasound is an effective way to detect meeting room boundaries due to the property that acoustic waves are more likely to be contained within a meeting room's walls than electromagnetic waves. Conversely, an electromagnetic signal, such as a BT or WiFi signal, may not be contained by the meeting room's walls, therefore it is not always possible to tell if an IHS (that can receive the electromagnetic signal) is located inside or outside a meeting room.

(130) FIG. **7** is a diagram illustrating an example of architecture **700** usable for handling conference room boundaries and/or context. As shown, architecture **700** includes certain components disposed in conference room **100** and others residing in cloud **504**.

(131) In this example, conference room **100** includes client IHSs **104**A-N in communication with video bar **101** and/or host IHS **105**A (client IHSs B-N are not shown). Each of IHSs **104**A-N may be configured to execute a respective client IHS agent **510**A-N. Video bar **101** and/or host IHS **105**A may also include, or otherwise be coupled to, display(s) **102**A-N, audio device(s) **306**/**207**, camera(s) **308**, digital whiteboard **509**, and touch display or controller **105**B. In other examples, however, other devices may be present in room **100** and one or more of the devices shown may be absent.

(132) In room **100**, video bar **101** and/or host IHS **105**A are configured to implement, execute, or instantiate OTB agent **501** and context service **701**. In addition to agent **510**A, client IHS **104**A is also configured to implement, execute, or instantiate execution engine **702**.

(133) On cloud **504**, peripheral management service **503** is coupled to device and configuration database **505**. Peripheral management service **503** may be configured to use UC API **704** to communicate with a UC service via UC application integrator module **703**, and use room reservation API **706** to communicate with a room reservation, calendar, or scheduling service via room reservation integrator module **705**.

(134) Context service **701** may be configured to periodically collect and/or broadcast a video bar's ID, a meeting ID, and a temporary meeting session key using an ultrasonic signal or beacon via speaker(s) **307**, as orchestrated by OTB agent **501**. To that end, context service **701** may communicate with peripheral management service **503** to retrieve, for example, meeting information such as a meeting ID, a meeting application, the duration of a meeting, invited participants, present participants, role(s) of IHS user(s), etc. (e.g., from a UC service via UC API **704**), and room reservation information such as: a room name or ID, a device list, a virtual map of the room, wireless credentials (usable to establish subsequent BT or WiFi connections, etc.), and so on (e.g., from a reservation, calendar, or scheduling service via room reservation API **706**).

(135) When a user of IHS **104**A enters room **100**, client IHS agent **510**A may receive the ultrasonic signal or beacon, retrieve the meeting ID from the signal, and verify it against US application information, calendar or email invites, etc. Once verified, a meeting room context may be established with room data and meeting data.

(136) After the meeting room context is established, execution engine **702** may perform one or

more operations that modify the user's experience based on the context, and/or it may present those options to the user (e.g., via a GUI). Such operations may include, but are not limited to: muting/unmuting meeting room microphones, connecting to an in-room wireless display, automatically joining a collaboration session with participants in the same meeting room, etc.

(137) When the user leaves room **100**, IHS **104**A stops receiving the ultrasonic broadcast. In response, execution engine **702** may remove or purge the context, such that the options provided by execution engine **702** are no longer available.

(138) As such, in some cases, method **700** may be used to take attendance of participants who are in room **100** during a remote session (e.g., by periodically re-broadcasting the ultrasonic signal, etc.).

(139) In other cases, if video bar **101** and/or host IHS **105**A determines that IHS **104**A is near (and/or in the direction of) a door or window, video bar **101** may change the amplitude of the ultrasonic signal. For example, by reducing the amplitude of the ultrasonic signal containing the meeting room context, IHS **104**A may only receive the signal it if is closer to video bar **101** than other IHSs that are not near (and/or in the direction of) the door or window, in which case IHS **104**A is more likely to be within room **100**.

(140) FIGS. **8**A and **8**B are diagrams illustrating examples of methods **800**A and **800**B for handling conference room boundaries and/or context. In various embodiments, methods **800**A and **800**B may be performed, at least in part, by components of architecture **700** of FIG. **7**.

(141) Method **800**A begins at **801**, where video bar **101** transmits ultrasonic signals containing meeting room context information within meeting room **100**. These ultrasonic signals may also include a list of parameters that the user may control in room **100**, including, but not limited to, microphone mute/unmute, lighting on/off, connect to a wireless display, etc.

(142) In some implementations, ultrasonic signals may be transmitted with an amplitude such that, after undergoing acoustic spreading and absorption, are not detectable by IHS disposed outside the boundaries of room **100**. In other cases, once the location of a given IHS in room **100** is determined (e.g., using method **600**), a speaker array may specifically direct the ultrasonic signal toward the given IHS.

(143) At **802**, a user enters meeting room **100**. At **803**, the user's IHS (e.g., IHS **104**A) receives the in-room broadcast via an ultrasonic signal or other proximity technologies. At **804**, IHS **104**A verifies whether the meeting ID of an ongoing or upcoming meeting matches a meeting recorded in the user's calendar or email applications. It not, method **800**A ends at **805**.

(144) Otherwise, at **806**, IHS **104**A may provide or render a control GUI to the user with a selectable list of parameters or options, such as those included in the ultrasonic signal. At **807**, the user selects one or more options, which sends a corresponding message to OTB agent **501**. For example, the user may select a video cast option whereby IHS **104**A derives a WiFi hostname from the meeting room context, connects to a wireless display, and starts a screencast session.

(145) At **808**, OTB agent **501** sends a command to a room resource to implement the user's selection (e.g., an Audio Digital Signal Processor or “ADSP,” a graphics or controller, a room lighting controller, etc.).

(146) Method **800**B begins at **809**, upon the user leaving room **100**, for example, as detected using method **600**. At **810**, IHS **104**A no longer receives the ultrasonic broadcast until a time-out event. In response, at **811**, client IHS agent **510**A purges the current meeting room context, at least in part, by invalidating the temporary meeting session key, so that the user is no longer able to control room resources. Method **800**B ends at **812**.

(147) In some cases, an attendance policy may determine when (e.g., how often, contextual triggers, etc.) and how (e.g., by ultrasound) to take attendance or otherwise record the presence or absence of participants during a remote meeting.

(148) In many situations, however, IHSs may provide their users with controls to mute (and unmute) its integrated microphone (e.g., **114**A), for example, for privacy or security purposes (e.g.,

to prevent conversation-related information from inadvertent or malicious dissemination). Ordinarily, if the IHS's microphone is muted, methods **800**A and **800**B would not work because the IHS would not be able to receive the video bar **101**'s ultrasonic broadcast.
(149) To address this, and other concerns, in some cases, in response to detecting a BT signal in the room (e.g., transmitted at **601** in FIG. **6**), the ultrasonic signal may be received by an IHS regardless of whether the IHS's microphone is muted or not, without loss of privacy. For instance, if an IHS's microphone is muted, instead of turning or leaving the microphone off, client IHS agent **510**A may leave the audio input channel, on and it may apply a high-pass filter to the captured signal configured to select ultrasonic frequencies to the exclusion of audible frequencies.
(150) FIGS. **9**A and **9**B are diagrams illustrating examples of methods **900**A and **900**B for handling conference room boundaries and/or context when a microphone is muted. In various embodiments, methods **900**A and **900**B may be performed, at least in part, by components of architecture **700** of FIG. **7**.
(151) Method **900**A begins at **901**, where video bar **101** transmits an advertising signal, such as a BLE ADV signal (scannable, non-directed, and non-connectable) indicating the presence of an ultrasonic broadcast in room **100**. At **902**, client IHS agent **510**A determines if the client IHS's microphone is muted or turned off.
(152) At **903**, client IHS agent **510**A determines if the IHS's microphone may be used. For example, the user may be able to set a policy or setting to determine whether the client IHS's microphone may be temporarily enabled to receive ultrasound signals (when the client IHS's microphone is originally muted or turned off), and **903** client IHS agent **510**A checks that policy.
(153) At **904**, if the policy allows, client IHS **104**A may listen for the BLE ADV signal. In response to its detection at **905**, client IHS **104**A may turn the microphone on (or unmute it) temporarily using execution engine **702**. At **906**, If BLE ADV is detected by the IHS, a SCAN_REQ is sent, triggering video bar **101** to send a SCAN_RESP with the room or meeting ID.
(154) At **906**, if no policy has been set, method **900**B may prompt the client IHS user to manually turn on the client IHS's microphone to perform meeting room detection, and it may apply a high-pass filter (e.g., with a cutoff frequency ˜20 kHz) to the audio input channel.
(155) At **907**, the client IHS may listen for the ultrasonic broadcast containing a room or meeting ID. At **908**, after having received the ultrasonic signal, the client IHS **104**A may turn off (or mute) the microphone once again (e.g., via execution engine **702**) before method **900**A ends at **909** after a defined duration.
(156) Method **900**B begins at **910**. At **911**, video bar **101** transmits the BLE ADV signal indicating the presence (current or impending) of the ultrasonic broadcast in room **100**. Then, in response to having received the SCAN_REQ of **906**, at **912** video bar **101** broadcasts the ultrasonic signal with the meeting room context before method **900**B ends at **913**.
(157) As such, these systems and methods may identify, collect, manage, and remove meeting room context across video bar **101**, IHSs **104**A-N, and other devices in connection with ongoing or upcoming remote meetings. In various implementations, these systems and methods may offer easy-to-use features that automate selected operations using the information received as meeting room context.
(158) It should be noted that, in various embodiments, architecture **700** may be combined with components of other architectures described herein, and methods **800**A, **800**B, **900**A, and **900**B may be combined with operations of other methods described herein, to provide additional or alternative features.
(159) As described herein, in an illustrative, non-limiting embodiment, a video bar may include a processor and a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution by the processor, cause the video bar to: transmit an ultrasonic signal in a conference room, where the ultrasonic signal comprises at least one of: room data, meeting data, or a peripheral setting; and establish or maintain a connection with a client IHS, at least in

part, in response to a message from the client IHS indicating receipt of the ultrasonic signal.
(160) The room data may include an indication of at least one of: a room identifier, a list of devices present in the conference room, or a wireless credential. The program instructions, upon execution by the processor, cause the video bar to determine that a subsequent message from the client IHS has not been received in response to a subsequent ultrasonic signal and, in response to the determination, terminate the connection. To terminate the connection, the program instructions, upon execution by the processor, cause the video bar to invalidate the wireless credential.
(161) Prior to the termination of the connection, the program instructions, upon execution by the processor, cause the video bar to request a user of the client IHS to provide an access code. The message may be received via: (i) an electrical or electromagnetic communication channel, or (ii) an ultrasonic communication channel. Moreover, the message may include an indication of the signal strength of the ultrasonic signal as received by the client IHS.
(162) The program instructions, upon execution by the processor, cause the video bar to establish or maintain the connection, at least in part, in response to a determination that the signal strength meets or exceeds a threshold value. The determination may be performed, in part, in response to another determination that a location of the client IHS matches a direction of a door or window in the conference room.
(163) The program instructions, upon execution, may cause the video bar to terminate the connection, at least in part, in response to a determination that the signal strength does not meet or exceed the threshold value or another threshold value.
(164) The meeting data may include an indication of at least one of: a meeting identifier, a meeting application, a meeting duration, invited participants, or participants present in the conference room. The program instructions, upon execution by the processor, cause the video bar to determine, based at least in part upon one or more messages received from the client IHS, whether the meeting data matches calendar information of a user of the client IHS.
(165) Also, the program instructions, upon execution by the processor, cause the video bar to terminate the connection, at least in part, in response to a determination that the meeting data does not match the calendar information.
(166) The peripheral setting may include an indication of at least one of: a microphone mute/unmute setting, a room lighting setting, a shared display setting, or a shared camera setting. The client IHS may be configured to apply the peripheral setting to at least one of a plurality of peripheral devices disposed in the conference room.
(167) The processor may be coupled to a host IHS external to the video bar, and to establish or maintain the connection, the program instructions, upon execution, cause the video bar to request the host IHS to establish or maintain the connection. The video bar or host IHS may be configured to maintain a virtual map of one or more client IHSs disposed in the conference room based, at least in part, upon the message.
(168) In another illustrative, non-limiting embodiment, a hardware memory device may have program instructions stored thereon that, upon execution by a processor of a client IHS, cause the client IHS to: receive an ultrasonic signal from a video bar in a conference room, where the ultrasonic signal comprises a session key and the video bar is coupled to a host IHS; and in response to the ultrasonic signal, use the session key to establish or maintain a data connection with the video bar or the host IHS.
(169) The program instructions, upon execution by the processor, may further cause the client IHS to a invalidate the session key in response to not having received a subsequent ultrasonic signal.
(170) In yet another illustrative, non-limiting embodiment, a method may include: determining, at least in part using a video bar, a presence of an IHS with respect to a conference room; and at least one of: requesting an access code from a user of the IHS, at least in part, in response to the IHS being outside the conference room, or not requesting the access code from a user of the IHS, at least in part, in response to the IHS being inside the conference room.

(171) In many situations, video bar **101** may connect to external devices, for example, to achieve broader coverage (e.g., additional display, additional camera/mic, etc.) and/or to allow a user to bring their personal IHS or peripherals for participating in a remote meeting (e.g., for presenting a video/deck). Ordinarily, however, these device additions are stateless and undergo no security checks, which creates security concerns (e.g., malware attacks, unauthorized device access, etc.).

(172) Using systems and methods described herein, however, only known and trusted devices may be added to a conference room, and all other devices may be rejected. Additionally, or alternatively, these systems and methods may allow a device to connect to video bar **101** for a specific remote meeting, for a selected duration only (e.g., the reserved meeting time). In cases where the user brings their own IHS, these systems and methods may allow basic access (e.g., using an HDMI port, etc.) while other features are restricted (e.g., wireless connections).

(173) FIG. **10** is a diagram illustrating an example of architecture **1000** usable for securely adding devices to a conference room. As shown, architecture **1000** includes certain components disposed in conference room **100** and others residing in cloud **504**.

(174) In this example, conference room **100** includes client IHSs **104**A-N in communication with video bar **101** and/or host IHS **105**A (client IHSs B-N are not shown). Each of IHSs **104**A-N may be configured to execute a respective client IHS agent **510**A-N. Video bar **101** and/or host IHS **105**A may also include, or otherwise be coupled to, display(s) **102**A-N, audio device(s) **306**/**207**, camera(s) **308**, digital whiteboard **509**, and touch display or controller **105**B. In other examples, however, other devices may be present in room **100** and one or more of the devices shown may be absent.

(175) In room **100**, video bar **101** and/or host IHS **105**A are configured to implement, execute, or instantiate OTB agent **501** and upper filter driver **1001**. Upper filter driver **1001** sits above a primary driver in the driver stack. In this case, upper filter driver **1001** may be configured to ensure that selected devices (e.g., all devices) have at least a subset of their functionality and/or features blocked (e.g., upon entering room **100**). OTB agent **501** allows video bar **101** and/or host IHS **105**A to communicate with peripheral management service **503** to receive information necessary to perform its operations. Meanwhile, IHS **104**A is configured to implement, execute, or instantiate client IHS agent **510**A.

(176) On cloud **504**, peripheral management service **503** is coupled to device and configuration database **505**, room booking services **1002**, and AuthZ and AuthN service(s) **507**, which in turn are coupled to user database **508**. Room booking services **1002** may enable users to schedule remote meetings, and it may keep a calendar with meeting details (e.g., meeting ID, time, duration, list of participants or attendees, required or optional participation for each attended, the role of each participant in the meeting, whether the meeting is expected to be confidential or privileged, whether session recording and/or transcription are allowed, a conference room ID, a location of the room, etc.).

(177) Moreover, peripheral management service **503** may connect with AuthZ and AuthN service(s) **507** to access user information stored in user database **508**. Additionally, peripheral management service **503** may maintain a list of trusted devices (e.g., IHSs that are part of the same corporate/enterprise) in device and configuration database **505**.

(178) FIG. **11** is a diagram illustrating an example of method **1100** for securely adding devices to a conference room. In various embodiments, method **1100** may be performed, at least in part, by components of architecture **1000** of FIG. **10**.

(179) Method **1100** begins at **1101**, where peripheral management service **503** receives IT administrator policies. At **1102**, peripheral management service **503** establishes a session with OTB agent **501** of video bar **101** and/or host IHS **105** based, at least in part, upon those policies.

(180) Upon initialization, OTB agent **501** enumerates all connected devices but blocks their functionality and/or features using upper filter driver **1001** (e.g., by default). Particularly, at **1103**, OTB agent **501** receives each of IHSs **104**A-N's (and other devices') serial number, service tag, or

other identifying information and enumerates those devices. At **1104**, OTB agent **501** sends a command to upper filter driver **1101** to place the enumerated devices in blocked mode, and at **1105** upper filter driver **1101** executes the command to block selected features or functionality in the enumerated devices with respect to room **100**.
(181) At **1106**, OTB agent **501** sends a device detail list to peripheral management service **503**, and at **1107** peripheral management service **503** checks, against device and configuration database **505**, if the devices on the list are trusted, for how long (e.g., trust time equal to the duration of a meeting, trusted for a day, etc.), and/or whether under a set of conditions or restrictions (e.g., allowed or forbidden features based upon contextual information, such as, for example, location of room **100**, identity of users, type of remote meeting, participants of the meeting, etc.).
(182) At **1108**, peripheral management service **503** sends these trust details, for each device on the list, to OTB agent **501**. To reduce delays, a list of previously trusted devices list may be cached. At **1109**, for each device on the list, if a given device is trusted, OTB agent **501** sends a command to upper filter driver **1101** to unblock previously blocked functionality and/or features of the given device(s). At **1110**, upper filter driver **1101** unblocks the functionality and/or features of the given device(s).
(183) At **1111**, if the trust of a given device is timed, OTB agent **501** may cache the device's details, and, upon expiration of the trust time, it may request upper filter driver **1101** to block the device (or a subset of its functionality and/or features) again. At **1112**, method **1100** loops back to **1104** for any new devices connected to video bar **101** and/or host IHS **105**A.
(184) In some cases, if a device is not trusted to be permanently part of the conferencing room, but the user wants to use it, method **1100** may verify that the connected device is included in the user's device list, and that the user is part of an ongoing or upcoming remote meeting. If so, the user's device may be allowed to connect only for the duration of the session. Moreover, if the device is not a corporate device (e.g., Bring Your Own Device or "BYOD"), then only the reduced functionality and features may be allowed.
(185) As such, systems and methods described herein provide the secure addition of one or more devices (both smart and non-smart devices) in a conferencing room solution. Moreover, these systems and methods provide the ability to scale for both corporate devices and BYODs.
(186) It should be noted that, in various embodiments, architecture **1000** may be combined with components of other architectures described herein, and method **1100** may be combined with operations of other methods described herein, to provide additional or alternative features.
(187) As described herein, in an illustrative, non-limiting embodiment, a video bar may include a processor and a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution by the processor, cause the video bar to: prevent a client IHS from using a feature available in a conference room; and allow the client IHS to use the feature in response to a determination that at least one of: (i) the client IHS, or (ii) a user of the client IHS, is expected to be in the conference room.
(188) The program instructions, upon execution, cause the video bar to prevent the client IHS from using the feature in response to having received a message, from the client IHS, comprising an identification of at least one of: (i) the client IHS, or (ii) the user of the client IHS. For example, the identification may include at least one of: a serial number, or a service tag.
(189) To determine that the client IHS or the user is expected to be in the conference room, the program instructions, upon execution by the processor, cause the video bar to: compare the identification against a list of attendees of an ongoing or upcoming remote conferencing session, or receive results of the comparison from a cloud service. Moreover, to prevent the client IHS from using the feature, the program instructions, upon execution by the processor, may further cause the video bar to execute an upper-level filter driver.
(190) The feature may include a feature of a peripheral device disposed in the conference room. The peripheral device may include at least one of: a shared display, a shared microphone, a shared

speaker, or a shared camera.
(191) Additionally, or alternatively, the feature may include a remote conferencing application feature. For example, the remote conferencing application feature may enable the client IHS to join a remote conferencing session. Additionally, or alternatively, the remote conferencing application feature may enable the client IHS to participate in one or more aspects of a remote conference session. For example, the one or more aspects may include: receiv(ing) video, transmit(ting) video, receiv(ing) audio, transmit(ting) audio, or shar(ing) an electronic file. Additionally, or alternatively, the remote conferencing application feature may enable recording of one or more aspects of the remote conferencing meeting.
(192) To allow the client IHS to use the feature, the program instructions, upon execution, cause the video bar to allow the client IHS to use the feature for a selected amount of time.
(193) The program instructions, upon execution by the processor, may cause the video bar to select the amount of time based, at least in part, upon an identification of at least one of: (i) the client IHS, or (ii) the user. Additionally, or alternatively, the program instructions, upon execution by the processor, cause the video bar to select the feature based, at least in part, upon an identification of at least one of: (i) the client IHS, or (ii) the user. Additionally, or alternatively, the program instructions, upon execution by the processor, cause the video bar to select the feature based, at least in part, upon a policy received from an Information Technology Decision Maker (ITDM).
(194) In another illustrative, non-limiting embodiment, a hardware memory device may have program instructions stored thereon that, upon execution by a processor of a client IHS, cause the client IHS to: transmit an identification of at least one of: (i) the client IHS, or (ii) a user of the client IHS, to a video bar disposed in a conference room, where the video bar is configured to prevent the client IHS from using a device available in a conference room; and in response to the video bar having authenticated or authorized the identification, access the device. To prevent the client IHS from using the device, the video bar may be configured to execute an upper-level filter driver.
(195) In yet another illustrative, non-limiting embodiment, a method may include: blocking a client IHS from using one or more peripheral devices available in a conference room; validating an identification of at least one of: (i) the client IHS, or (ii) a user of the client IHS, received by a video bar disposed in the conference room; and in response to the validation, allowing the client IHS to use at least a feature of the one or more peripheral devices. The feature may be selected, at least in part, based upon the identification.
(196) In various embodiments, systems and methods described herein may integrate conference room solutions (e.g., video bar **101** and/or host IHS **105**A) with the room booking services. These systems and methods may also implement zero-touch or seamless authentication mechanisms to ensure that users expected to participate in a remote meeting are in room **100** during the remote meeting, and/or that users not expected to participate in the remote meeting are not in room **100** during the remote meeting.
(197) Specifically, using systems and methods described herein, video bar **101** and/or host IHS **105**A may ensure that only a remote meeting's host or organizer and invited attendees are present in room **100** during the remote meeting. These systems and methods may authenticate meeting room users without requiring additional operations, such as tapping a badge or entering a password or code, so that the right user(s) can seamlessly use room **100**. Video bar **101** and/or host IHS **105**A may also prevent users who are not part of the meeting from barging in.
(198) Depending upon a user's identification and/or other user information such as, for example: the role of the user in a remote meeting (e.g., as host/organizer or participant, as a presenting or non-presenting participant, as employee or contractor, etc.), the user's job rank or position in the enterprise (e.g., executive, manager, engineer, IT administrator, etc.), and so on, these systems and methods may selectively enable features within room **100** for that user.
(199) In some cases, users may be associated with a group, and the aforementioned selective

enablement of features may be performed based on the users' group, for every user in that group. Examples of features include, but are not limited to, access to peripheral devices (e.g., external displays, cameras, whiteboard, etc.), specific device features, UC application features (e.g., start or stop recording a session, share documents or GUI windows during the session, etc.), and so on.

(200) FIG. **12** is a diagram illustrating an example of architecture **1200** usable for identifying and authenticating users in a conference room. As shown, architecture **1200** includes certain components disposed in conference room **100** and others residing in cloud **504**.

(201) In room **100**, video bar **101** and/or host IHS **105**A are configured to implement, execute, or instantiate OTB agent **501** and upper filter driver **1001**. OTB agent **501** allows video bar **101** and/or host IHS **105**A to communicate with peripheral management service **503**. Meanwhile, IHS **104**A is configured to implement, execute, or instantiate client IHS agent **510**A. In this environment, digital whiteboard **509** is also present in room **100** and coupled to video bar **101** and/or host IHS **105**A.

(202) On cloud **504**, peripheral management service **503** is coupled to gateway **1201** and to device and configuration database **505**, conference room security service **1202**, and AuthZ and AuthN service(s) **507**, which in turn are coupled to user database **508**.

(203) Gateway **1201** may be configured to expose various public APIs to devices located in room **100**, and to perform conventional gateway operations (e.g., reverse proxy, load balancing, rate limiting, filtering traffic, etc.). Conference room security service **1202** may be configured to identify and verify users in room **100** along with AuthZ and AuthN service(s) **507**.

(204) FIG. **13** is a diagram illustrating an example of method **1300** for identifying and authenticating users in a conference room. In various embodiments, method **1300** may be performed, at least in part, by components of architecture **1200** of FIG. **12**.

(205) Method **1300** includes initialization phase **1301-1306**, followed by steady state **1307-1319**. Particularly, method **1300** begins at **1301**, where peripheral management service **503** establishes a session with OTB agent **501** of video bar **101** and/or host IHS **105**, as well as conference room security service **1202**.

(206) At **1302**, OTB agent **501** enumerates all connected devices. At **1303**, OTB agent **501** registers the detected devices with peripheral management service **503**, and at **1304** it receives an acknowledgement. Then, at **1305**, OTB agent **501** sends information about the applications being used in room **100** and connected device capabilities and/or features to peripheral management service **503**. At **1306**, OTB agent **501** receives another acknowledgment before steady-state operation begins.

(207) During steady state, at **1307**, client IHS agent **510**A connects to video bar **101** and/or host IHS **105**A. At **1308**, client IHS agent **510**A establishes a session with peripheral management service **503** and conference room security service **1202**. At **1309**, OTB agent **501** blocks data and device access in room **100**, for example, through upper filter driver **1101**.

(208) At **1310**, client IHS agent **510**A may get a token (e.g., a JSON Web Token or “JWT,” a Security Assertion Markup Language or “SAML” token, etc.), for example, from AuthZ and AuthN service(s) **507** (or some other identity management service), and at **1311** it may create a registration request (e.g., a Hypertext Transfer Protocol or “HTTP” request) containing the token (e.g., in an HTTP header).

(209) At **1312**, client IHS agent **510**A sends the registration request to peripheral management service **503**. At **1313**, peripheral management service **503** extracts the token from the request and sends it to conference room security service **1202**. At **1314**, conference room security service **1202** validates the token and retrieves a corresponding user ID, for example, from AuthZ and AuthN service(s) **507**.

(210) At **1315**, conference room security service **1202** sends the user ID to peripheral management service **503**. At **1316**, peripheral management service **503** verifies, based upon the user ID, whether the user is valid and checks an attendee list, via room booking services **1003**, for an ongoing or upcoming remote meeting. If the verification is successful, at **1317** peripheral management service

503 may send notifications to client IHS agent **510**A and to OTB agent **501**.
(211) At **1318**, OTB agent **501** unblocks the user's device (or eliminates/modifies feature restrictions imposed in **1311**) and allows a data path to be established. The selection of which features to enable may be made based, at least, in part, upon the user ID, the role of the user in the remote meeting, the user's job or position in an enterprise, etc. For example, in some cases, a user who is a company executive may have access to a shared camera **308** (e.g., a Pan, Tilt, and Zoom or "PZT" camera), but whiteboard **509** is disabled, whereas another user who is an engineer may have access to whiteboard **509** while access to shared camera **308** is disabled. At **1319**, method **1300** loops back to **1315** to process additional or newly connected devices.
(212) As such, systems and methods described herein may enable video bar **101** and/or host IHS **105**A to seamlessly (zero-touch) identify and authenticate a user and their IHS or devices upon connecting to room **100**, and to selectively allow or block access to in-room devices and application features based upon the user. As with other systems methods described herein, these techniques may be scaled across different video bar and/or host IHS **105**A's OSs, client IHS's OSs, and identity management systems (e.g., AZURE, AMAZON WEB SERVICES, etc.).
(213) It should be noted that, in various embodiments, architecture **1200** may be combined with components of other architectures described herein, and method **1300** may be combined with operations of other methods described herein, to provide additional or alternative features.
(214) In an illustrative, non-limiting embodiment, a video bar may include a processor and a memory coupled to the processor, the memory having program instructions stored thereon that, upon execution by the processor, cause the video bar to: prevent a client IHS from using a feature available in a conference room; and allow the client IHS to use the feature in response to a determination that: (i) a token provided by client IHS is valid, and (ii) a user of the client IHS is a participant of an ongoing or upcoming conference session.
(215) The program instructions, upon execution, cause the video bar to prevent the client IHS from using the feature in response to having received a message, form the client IHS, comprising an identification of the client IHS. The identification may include at least one of: a serial number, or a service tag.
(216) To determine that the user of the client IHS is a participant of an ongoing or upcoming conference session, the program instructions, upon execution by the video bar, cause the video bar to: retrieve an identification of the user based, at least in part, upon the token; and compare the identification of the user against a list of attendees of the ongoing or upcoming conference session. The token may be provided by the client IHS as part of an HTTP request.
(217) The program instructions, upon execution, cause the video bar to allow the client IHS to use the feature at least in part, upon a determination that the user has a presenter, host, or organizer role in the ongoing or upcoming conference session. Additionally, or alternatively, the program instructions, upon execution, cause the video bar to allow the client IHS to use the feature at least in part, upon a determination that the user has a non-presenting participant role in the ongoing or upcoming conference session. Additionally, or alternatively, the program instructions, upon execution, cause the video bar to allow the client IHS to use the feature at least in part, upon a determination that the user belongs to a selected group or category of users the ongoing or upcoming conference session.
(218) To prevent the client IHS from using the feature, the program instructions, upon execution by the processor, further cause the video bar to execute an upper-level filter driver. For example, the feature may include a feature of a peripheral device disposed in the conference room. The peripheral device may include at least one of: a shared display, a shared microphone, a shared speaker, or a shared camera.
(219) The feature may include a remote conferencing application feature. The remote conferencing application feature may enable the client IHS to join a remote conferencing session. Additionally, or alternatively, the remote conferencing application feature may enable the client IHS to

participate in one or more aspects of a remote conference session. The one or more aspects may include: receiv(ing) video, transmit(ting) video, receiv(ing) audio, transmit(ting) audio, or shar(ing) an electronic file. Additionally, or alternatively, the remote conferencing application feature may enable recording of one or more aspects of the remote conferencing meeting.

(220) To allow the client IHS to use the feature, the program instructions, upon execution, cause the video bar to allow the user to use the feature for a selected amount of time. The program instructions, upon execution by the processor, cause the video bar to select the amount of time based, at least in part, upon a role of the user in the ongoing or upcoming conference session.

(221) In another illustrative, non-limiting embodiment, a hardware memory device may have program instructions stored thereon that, upon execution by a processor of a client IHS, cause the client IHS to: transmit a token from a client IHS to a video bar disposed in a conference room, where the video bar is configured to prevent the client IHS from accessing one or more features of the conference room, and where the token is usable to identify a user of the client IHS; and in response to the video bar having authenticated or authorized the token, access the one or more features depending upon a role of the user in an ongoing or upcoming conference session.

(222) In yet another illustrative, non-limiting embodiment, a method may include: blocking a client IHS from using one or more conferencing features during an ongoing or upcoming conference session; validating a token received by a video bar disposed in the conference room, where the token is associated with a user of the client IHS; in response to the validation, identify a role of the user in the ongoing or upcoming conference session; and in response to the validation, allowing the client IHS to access the one or more conferencing features in accordance with the role of the user.

(223) Furthermore, some embodiments may provide for managing bring your own device (BYOD) device entitlements in a conferencing room. In this example, a BYOD device includes a portable computing device (e.g., laptop or smart phone) that does not have an enterprise-approved image installed. In other words, the BYOD device is not enterprise-issued but, rather, may be a personal device of a user. It may be desirable in some instances to allow such devices to have convenient access to conference rooms.

(224) In some conferencing room solutions, there may be multiple ways for users to participate, such as using only the installed conferencing room device for the meeting (e.g., video bar **101**), using a corporate-issued and trusted device in tandem with the conferencing room device, and/or using a BYOD device in tandem with the conferencing room device. However, there is currently no seamless solution available to enable features and entitlements of a BYOD user as a result of device compatibility, security posture, and user group issues.

(225) For instance, an end user may desire to have access to some or all of the conferencing room features with minimal steps and ease of setup. Information technology (IT) administrators may desire to enable the features for BYOD users but may require security identification and group type identification before the BYOD device may fully participate. As an example, an IT administrator may desire to enable a set of features only after identifying a device's trust level so that the conference room device and the solution is secure. Further in the example, an IT administrator may desire to enable the set of features for a BYOD user after successful identification, where perhaps different users may have different features enabled.

(226) FIG. **14** is a diagram illustrating an example of architecture **1400** usable for managing BYOD devices and users in a conference room. As shown, architecture **1400** includes certain components disposed in conference room **100** and others residing in cloud **504**. For purposes of the present example, it will be assumed that IHS **104**A is a BYOD device, and that the other IHSs **104** may be enterprise-issued trusted devices. However, the scope of implementations may facilitate any number of BYOD devices as appropriate.

(227) In the example of FIG. **14**, the OTB agent **501** is installed in the conferencing room device/controller/host (e.g., video bar **101**). The OTB agent **501** may establish a session with peripheral management service **503**, pull a trusted BYOD device list from the peripheral

management service (e.g., which may be stored at either or both of databases **505**, **508**). The IHS agent **510**A, installed on the IHS **104**A, in this example is configured as a BYOD agent.
(228) The BYOD agent functionality may be a web application or similar application that is loaded to the IHS **104**A through a universal resource locator (URL) or quick response (QR) code that is shared in a trusted manner or displayed in the conferencing room. The BYOD agent functionality may use regular, unprivileged APIs to communicate to the peripheral management service **503** and, thus, not employ administrative access for its execution. The communication between the client IHS agent **510**A and the peripheral management service **503** may allow the peripheral management service **503** to acquire user and device details from client IHS **104**A, such as operating system, hostname, device name, MAC address, IP address, and the like. Such information may allow the peripheral management service **503** to identify client IHS **104**A by its hardware and software, and peripheral management service **503** may further save such details in either or both of databases **505**, **508** in case of future access by the same device.
(229) Additionally in this example, the peripheral management service **503** may communicate with the UC application API **704** service to acquire the meeting and participant details and also communicate with the authorization services **507** to verify and validate user authenticity. For instance, the UC application API **704** service may gather meeting and user information, such as meeting ID, conference room ID, user ID, IDs of other participants, and the like.
(230) In an example embodiment of operation of the system of architecture **1400**, the various components provide user and trust identification of BYOD devices. For instance, in response to a user connecting a BYOD device (e.g., IHS **104**A), by default the OTB agent **501** may block the data access by the IHS **104**A to the host IHS **105**A. The agent **510**A then sends a notification to the peripheral management service **503** regarding the connection. In response, the peripheral management service **503** may acquire the device and user details from the agent **510**A as well as meeting details from UC application API **704**. Peripheral services agent **503** may further access participant IDs from booking room services **1002** if desired. Based on this information, the peripheral management service **503** may determine, as explained in more detail with respect to FIGS. **16**A-B, to either allow or block BYOD access by the IHS **104**A to the host IHS **105**A.
(231) Continuing with this example, the architecture **1400** may also manage various feature entitlements for BYOD users. For instance, it may be desired to provide differentiated features to different users, such that some user classes receive different features than other user classes. In one example, peripheral management service **503** acquires user and group information via authentication services **507**, software and firmware details of the IHS **104**A via agent **510**A, and provides a differentiated set of features to the IHS **104**A based on those details. An example of entitlements, corresponding to user groups, is shown in Table 1.
(232) TABLE-US-00001 TABLE 1 User BYOD Security User role Allowed Group Posture in meeting entitlements Manager A/V installed and all Organizer Allow all trusted applications (high) Executive A/V installed but not Participant Allow all except for all applications are digital whiteboard trusted (medium) Designer A/V not updated (low) Participant Allow only external monitor; block all other peripherals
(233) Therefore, the architecture **1400** may allow for a more convenient feature enablement for BYOD users through validating their trust. As explained in more detail with respect to FIGS. **16**A-B, trust validation may include communications with connected identity managers. The architecture **1400** may also provide managed, differentiated features based on user group and BYOD security posture.
(234) FIG. **15** is a diagram illustrating an example method **1500** for managing BYOD devices and users in a conference room. In various embodiments, method **1500** may be performed, at least in part, by components of architecture **1400** of FIG. **14**.
(235) In one example, the user of the BYOD device has initiated a meeting via a conferencing application, and the BYOD device has the conferencing application running thereon. At action

1502, the peripheral management service **503** and the OTB agent **501** establish a session. At action **1504**, the peripheral management service **503** acquires trusted device details, such as from device and configuration database **505**. However, in this example, the particular BYOD device in issue is not yet identified in the database **505**.

(236) At action **1506**, a user joins with a BYOD device. In response, the OTB agent **501** blocks data access of the BYOD device at action **1508**. Action **1508** may also include allowing a control path for the BYOD device. In response to the lack of data access, the agent **510**A (in this example, including BYOD agent functionality) sends a notification regarding its connection to the peripheral management service **503**. The peripheral management service **503** requests device details at action **1512**, and the agent **510**A responds with device details at action **1514**. Examples of device details may include operating system, hostname, device name, MAC address, IP address, and the like. The device details at action **1512** may also include user ID in some implementations. Although not explicitly shown in FIG. **15**, the peripheral management service **503** may save those device details to database **505** after trust has been established.

(237) At action **1516**, the peripheral management service **503** verifies trust of the device and makes a decision regarding whether to allow or block the BYOD device. An example of verifying trust and making a decision is described with respect to FIGS. **16**A-B (explained in more detail below). If the decision at action **1516** is to allow access by the BYOD device, the peripheral management service **503** notifies the OTB agent **501**. In some examples, the peripheral management service **503** may also query a database (e.g., database **508**) to determine a user group of the user and a set of service entitlements corresponding to the user group. The peripheral management service **503** may notify the OTB agent **501** of the service entitlements. The OTB agent **501** may then enable all appropriate data paths, peripherals, and service entitlements for the BYOD device at action **1518**.

(238) As a result of method **1500**, the system has allowed for feature enablement and validation of BYOD users. As the method **1500** began, the user of the BYOD device had started a meeting using a conferencing application running on the BYOD device. Once action **1518** has been performed, the user may be able to switch the meeting from the application running on the BYOD device to a larger screen managed by the host IHS **105**A, if desired. From the point of view of the user, they have brought their BYOD device to the conference room and have connected to the conference room hardware. Within a matter of seconds, the user has seen that the peripherals have been enabled and that there is an option to migrate the meeting from the conference application on the BYOD device to the host IHS **105**A. An advantage of such applications may be increased efficiency, such as by requiring fewer steps from the user of the BYOD device to be integrated into the conference room. Furthermore, security and efficiency may be increased by offloading the establishment of trust to devices in the cloud **504**.

(239) FIGS. **16**A-B is a diagram illustrating an example of method **1600** for validating BYOD devices in a conference room. In various embodiments, method **1600** may be performed, at least in part, by components of architecture **1400** of FIG. **14**. In this example, the actions are performed by peripheral management service **503**, authorization services **507**, and application API service **704**.

(240) At action **1601**, the peripheral management service **503** acquires a current meeting participant list by communicating with the application API service **704**, which at action **1602** fetches and sends the participant list. In the present example, the UC application API service **704** corresponds to the conferencing application under the name ZOOM, though the scope of implementations may include any conferencing application now known or later developed and with appropriate APIs.

(241) The participant list includes an indication of the user ID associated with the user of the BYOD device. For instance, a user ID may conform to the format “name@domain.com.” The IDs of the other participants may conform to the same format as well. Based on the user ID, the peripheral management service **503** determines whether the user of the BYOD device is in the list at action **1603**. For instance, a user ID may have been gathered at action **1512**, and action **1603** may include comparing the participant list to the user ID. Assuming that the user ID is in the list,

then the peripheral management service **503** may request the device ID of the BYOD device from the application API service **704**, which fetches and returns the requested information at action **1605**.

(242) At action **1606**, the peripheral management service **503** determines, based on the returned information, whether the device ID is in the trusted list of action **1504**. Assuming that the device is identified in the trusted list, then the peripheral management service **503** allows the device to participate at action **1607**.

(243) However, if the user is not in the list (action **1603**) or if the device is not in the trusted list (action **1606**), then the peripheral management service **503** communicates with the authorization service **507** to verify the user ID at action **1608**. The authorization service **507** may then send a response to the peripheral management service **503** that verification is either successful or failed at action **1609**. An example of authorization service **507** that may be used in some implementations is an active directory, such as the one provided in the cloud service having the name AZURE. For instance, authorization service **507** may serve the particular enterprise that manages conference room **100**, and it may serve other enterprises as well. The authorization service **507** may verify credentials for a multitude of different users across those enterprises.

(244) Assuming that verification is successful at action **1608**, then the peripheral management service **503** communicates with the authorization service **507** to notify the user to authenticate at action **610**. However, at actions **1608**, **1609**, if the user ID is not verified, then the verification has failed and the device participation is blocked at action **1612**. Similarly, if the user fails to authenticate at actions **1610**, **1611**, then the peripheral management service **503** may block the device from participating at action **1612**. However, assuming that the user is authenticated at actions **1610**, **1611**, then the successful validation leads to the peripheral management service **503** storing the user's device ID in the trusted device list (e.g., at database **505**) at action **1613** and allowing the device to participate at action **1614**.

(245) According to one non-limiting embodiment, a method includes identifying that a user device has connected to a conference room host device. For instance, the user device may include a laptop computing device or a tablet computing device that does not include an enterprise-approved image. More specifically, the user device may include a BYOD device. An example of a conference room host device may include host IHS **105**A. The connection between the user device and the conference room host device may include a wired or wireless connection, as appropriate.

(246) Continuing with the example, the method may further include blocking the user device from a data channel between a management agent at the conference room host device and the user device. An example of a management agent at the conference room host device may include OTB agent **501**. The method may further include receiving a message from a cloud-based management service to the management agent indicating an allowed status of the user device. An example of a cloud-based management service may include peripheral management service **503**. Once the conference room host device has received the message from the cloud-based management service, the conference room host device (or its agent) may open the data channel between the user device and the management agent.

(247) In a non-limiting embodiment, the method may further include disallowing access to a peripheral device by the user device prior to receiving the message indicating the allowed status. Once the message indicating the allowed status is received, the method may further include allowing access to the peripheral device by the user device. Furthermore, the method may include providing a prompt to a user of the user device, wherein the prompt is configured to allow the user to choose to migrate a conference session from the user device to the conference room host. The method may further include migrating the conference session in response to user input.

(248) In a non-limiting embodiment, the method may include receiving, from the cloud-based management service, an indication of a service entitlement associated with a user group of a user of the user device. An example of service entitlements and user groups is provided at Table 1. The

method may further include enabling the service entitlement in response to the indication of the service entitlement.

(249) In another non-limiting embodiment, a method may include receiving a notification from a user device, where the notification indicates that the user device has connected to a conference room host. In one example, the client agent **510**A may send a notification to peripheral management service **503** to indicate that the client IHS **104**A has connected to a conference room host device. The method may further include receiving user device details from the user device in response to a request for the user device details. The method may further include authenticating the user device based at least in part on the user device details and transmitting a message to the conference room host device indicating an allowed status of the user device.

(250) In a non-limiting embodiment, authenticating the user device may include acquiring a meeting participant list corresponding to a conference session that is active on the user device, determining that an entry in the participant list matches a user ID received with the user device details, and verifying the user ID according to a directory service. An example of a directory service may include authentication service **507**. The method may further include, prior to verifying the user ID, attempting to verify a device ID of the user device and verifying the user ID subsequent to failing to verify the device ID. The method may further include communicating with an interface of an application, corresponding to the conference session, including requesting and receiving the participant list from the application.

(251) In a non-limiting embodiment, authenticating the user device may include acquiring a meeting participant list corresponding to a conference session active on the user device, determining that an entry in the participant list does not match the user ID received with the user device details, and verifying the user ID according to a directory service.

(252) In a non-limiting embodiment, authenticating the user device may include acquiring a meeting participant list corresponding to a conference session active on the user device, determining that an entry in the participant list matches a user ID received with the user device details, acquiring a device ID of the user device, and verifying the device ID according to a trusted device list.

(253) In a non-limiting embodiment, the method may further include acquiring data associating the user of the user device with a user group and transmitting an indication of a service entitlement associated with the user group of the user device.

(254) A non-limiting embodiment may also include an IHS having one or more processors and a memory coupled to the one or more processors, wherein the memory comprises computer-readable instructions, which upon execution by the one or more processors, causes the IHS to perform the methods described above with respect to FIGS. **14-16**B. A further non-limiting embodiment may also include a non-transitory computer-readable storage device having instructions stored thereon for user device authentication, wherein execution of the instructions by one or more processors of an IHS causes the one or more processors to perform the methods described above with respect to FIGS. **14-16**B.

(255) To implement various operations described herein, computer program code (i.e., program instructions for carrying out these operations) may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, Python, C++, or the like, conventional procedural programming languages, such as the "C" programming language or similar programming languages, or any of ML/AI software. These program instructions may also be stored in a computer readable storage medium that can direct a computer system, other programmable data processing apparatus, controller, or other device to operate in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instructions which implement the operations specified in the block diagram block or blocks.

(256) Program instructions may also be loaded onto a computer, other programmable data

processing apparatus, controller, or other device to cause a series of operations to be performed on the computer, or other programmable apparatus or devices, to produce a computer-implemented process such that the instructions upon execution provide processes for implementing the operations specified in the block diagram block or blocks.

(257) Modules implemented in software for execution by various types of processors may, for instance, include one or more physical or logical blocks of computer instructions, which may, for instance, be organized as an object or procedure. Nevertheless, the executables of an identified module need not be physically located together but may include disparate instructions stored in different locations which, when joined logically together, include the module and achieve the stated purpose for the module. Indeed, a module of executable code may be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices.

(258) Similarly, operational data may be identified and illustrated herein within modules and may be embodied in any suitable form and organized within any suitable type of data structure. Operational data may be collected as a single data set or may be distributed over different locations including over different storage devices.

(259) Reference is made herein to "configuring" a device or a device "configured to" perform some operation(s). It should be understood that this may include selecting predefined logic blocks and logically associating them. It may also include programming computer software-based logic of a retrofit control device, wiring discrete hardware components, or a combination of thereof. Such configured devices are physically designed to perform the specified operation(s).

(260) It should be understood that various operations described herein may be implemented in software executed by processing circuitry, hardware, or a combination thereof. The order in which each operation of a given method is performed may be changed, and various operations may be added, reordered, combined, omitted, modified, etc. It is intended that the invention(s) described herein embrace all such modifications and changes and, accordingly, the above description should be regarded in an illustrative rather than a restrictive sense.

(261) Unless stated otherwise, terms such as "first" and "second" are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements. The terms "coupled" or "operably coupled" are defined as connected, although not necessarily directly, and not necessarily mechanically. The terms "a" and "an" are defined as one or more unless stated otherwise. The terms "comprise" (and any form of comprise, such as "comprises" and "comprising"), "have" (and any form of have, such as "has" and "having"), "include" (and any form of include, such as "includes" and "including") and "contain" (and any form of contain, such as "contains" and "containing") are open-ended linking verbs.

(262) As a result, a system, device, or apparatus that "comprises," "has," "includes" or "contains" one or more elements possesses those one or more elements but is not limited to possessing only those one or more elements. Similarly, a method or process that "comprises," "has," "includes" or "contains" one or more operations possesses those one or more operations but is not limited to possessing only those one or more operations.

(263) Although the invention(s) is/are described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention(s), as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention(s). Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

Claims

1. A method comprising: identifying that a user device has connected to a conference room host device; blocking the user device from a data channel between a management agent at the conference room host device and the user device in response to identifying that the user device has connected; receiving a message from a cloud-based management service to the management agent indicating an allowed status of the user device; and opening the data channel between the user device and the management agent in response to receiving the message indicating the allowed status.
2. The method of claim 1, further comprising: disallowing access to a peripheral device, connected to the conference room host device, by the user device prior to receiving the message indicating the allowed status; and allowing access to the peripheral device by the user device subsequent to receiving the message indicating the allowed status.
3. The method of claim 1, further comprising: providing a prompt to a user of the user device, wherein the prompt is configured to allow the user to choose to migrate a conference session from the user device to the conference room host device; and migrating the conference session in response to user input.
4. The method of claim 1, further comprising: receiving, from the cloud-based management service, an indication of a service entitlement associated with a user group of a user of the user device.
5. The method of claim 4, further comprising: enabling the service entitlement in response to the indication of the service entitlement.
6. The method of claim 1, wherein the user device comprises a laptop computing device or a tablet computing device, further wherein the laptop computing device or the tablet computing device does not include an enterprise-approved image.
7. The method of claim 1, wherein the method is performed by the conference room host device, and wherein the user device comprises a laptop computing device or a tablet computing device, further wherein identifying that the user device has connected includes detecting a wired connection between the user device and the conference room host device.
8. An Information Handling System (IHS), comprising: one or more processors; and a memory coupled to the one or more processors, wherein the memory comprises computer-readable instructions, which upon execution by the one or more processors, causes the IHS to: receive a notification from a user device, the notification indicating that the user device has connected to a conference room host device; receive user device details from the user device in response to a request for the user device details; authenticate the user device based at least in part on the user device details; and transmit a message to the conference room host device indicating an allowed status of the user device.
9. The IHS of claim 8, wherein the computer-readable instructions to cause the IHS to authenticate the user device comprises computer-readable instructions to cause the IHS to: acquire a meeting participant list corresponding to a conference session active on the user device; determine that an entry in the participant list matches a user ID received with the user device details; and verify the user ID according to a directory service.
10. The IHS of claim 9, further comprising computer-readable instructions to cause the IHS to: prior to verifying the user ID, attempt to verify a device ID of the user device; and verify the user ID subsequent to failing to verify the device ID.
11. The IHS of claim 9, wherein the computer-readable instructions to cause the IHS to acquire the meeting participant list comprises computer-readable instructions to cause the IHS to: communicate with an interface of an application, corresponding to the conference session, including requesting and receiving the participant list from the application.

12. The IHS of claim 8, wherein the computer-readable instructions to cause the IHS to authenticate the user device comprises computer-readable instructions to cause the IHS to: acquire a meeting participant list corresponding to a conference session active on the user device; determine that an entry in the participant list does not match a user ID received with the user device details; and verify the user ID according to a directory service.
13. The IHS of claim 8, further comprising computer-readable instructions to cause the IHS to: store a device ID of the user device in a trusted device list.
14. The IHS of claim 8, wherein the computer-readable instructions to cause the IHS to authenticate the user device comprises computer-readable instructions to cause the IHS to: acquire a meeting participant list corresponding to a conference session active on the user device; determine that an entry in the participant list matches a user ID received with the user device details; acquire a device ID of the user device; and verify the device ID according to a trusted device list.
15. The IHS of claim 14, further comprising computer-readable instructions to cause the IHS to: communicate with an interface of an application, corresponding to the conference session, including requesting and receiving the device ID from the application.
16. The IHS of claim 9, further comprising computer-readable instructions to cause the IHS to: transmit an indication of a service entitlement associated with a user group of a user of the user device.
17. A non-transitory computer-readable storage device having instructions stored thereon for user device authentication, wherein execution of the instructions by one or more processors of an IHS (Information Handling System) causes the one or more processors to: receive a notification from a user device, the notification indicating that the user device has connected to a conference room host device; receive user device details from the user device in response to a request for the user device details; authenticate the user device based at least in part on the user device details; and transmit a message to the conference room host device indicating an allowed status of the user device.
18. The non-transitory computer-readable storage device of claim 17, wherein the instructions to cause the one or more processors to authenticate the user device comprises instructions to cause the one or more processors to: acquire a meeting participant list corresponding to a conference session active on the user device; determine that an entry in the participant list matches a user ID received with the user device details; and verify the user ID according to a directory service.
19. The non-transitory computer-readable storage device of claim 18, further comprising instructions to cause the one or more processors to: prior to verifying the user ID, attempt to verify a device ID of the user device; and verify the user ID subsequent to failing to verify the device ID.
20. The non-transitory computer-readable storage device of claim 17, wherein the instructions to cause the one or more processors to authenticate the user device comprises instructions to cause the one or more processors to: acquire a meeting participant list corresponding to a conference session active on the user device; determine that an entry in the participant list does not match a user ID received with the user device details; and verify the user ID according to a directory service.